

Predictive Bank Client Deposit Rate Analysis

Portfolio Project

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Packages

```
library(dplyr)
library(ggplot2)
library(tidyr)
library(gmodels)
library(bestNormalize)
library(gridExtra)
library(caret)
library(pROC)
library(fastDummies)
```

Custom functions

```
create_plot <- function(data, var_name, type = "prop") {
  if (type == "count") {
    position <- "stack"
    colors <- c("TRUE" = "#1BC7E4", "FALSE" = "#E4381B")
  } else {
    position <- "fill"
    colors <- c("TRUE" = "#4CAF50", "FALSE" = "#F44336")
  }

  ggplot(data, aes(x = .data[[var_name]], fill = subscribed)) +
    geom_bar(position = position) +
    scale_fill_manual(values = colors) +
    labs(x = var_name, y = "", fill = "Subscribed") +
    theme_minimal() +
    theme(
      axis.text.x = element_text(angle = 45, hjust = 1),
      legend.position = "none"
    )
}

count_unknown <- function(data, value) {
  data %>%
    summarise(across(everything(), ~ sum(.x == value, na.rm = TRUE))) %>%
    pivot_longer(everything(), names_to = "column", values_to = "count") %>%
    filter(count > 0) %>%
    arrange(desc(count))
}

train_evaluate_model <- function(data) {
  set.seed(167)

  train_indices <- createDataPartition(data$subscribed, p = 0.8, list = FALSE)
  train_data <- data[train_indices, ]
  test_data <- data[-train_indices, ]

  model <- glm(subscribed ~ ., data = train_data, family = binomial)

  predictions <- predict(model, test_data, type = "response")
}
```

```
roc_curve <- roc(test_data$subscribed, predictions)
auc_value <- auc(roc_curve)

list(
  model = model,
  train_data = train_data,
  test_data = test_data,
  predictions = predictions,
  roc_curve = roc_curve,
  auc = auc_value
)
```

Importing data

```
bank_data <- read.csv("bankData/bank-full.csv", sep = ";")
```

Data cleaning

```
head(bank_data)
```

```
##   age      job marital education default balance housing loan contact day
## 1  58 management married  tertiary      no    2143     yes   no unknown   5
## 2  44 technician single  secondary      no     29     yes   no unknown   5
## 3  33 entrepreneur married secondary      no     2     yes  yes unknown   5
## 4  47 blue-collar married   unknown      no   1506     yes   no unknown   5
## 5  33      unknown single   unknown      no     1      no   no unknown   5
## 6  35 management married  tertiary      no    231     yes   no unknown   5
##   month duration campaign pdays previous poutcome y
## 1   may      261         1    -1         0 unknown no
## 2   may      151         1    -1         0 unknown no
## 3   may       76         1    -1         0 unknown no
## 4   may       92         1    -1         0 unknown no
## 5   may      198         1    -1         0 unknown no
## 6   may      139         1    -1         0 unknown no
```

```
tail(bank_data)
```

```
##      age      job marital education default balance housing loan
## 45206  25 technician single  secondary      no    505      no yes
## 45207  51 technician married  tertiary      no    825      no no
## 45208  71   retired divorced  primary      no   1729      no no
## 45209  72   retired married  secondary      no   5715      no no
## 45210  57 blue-collar married  secondary      no    668      no no
## 45211  37 entrepreneur married  secondary      no   2971      no no
##      contact day month duration campaign pdays previous poutcome y
## 45206 cellular  17  nov      386         2    -1         0 unknown yes
## 45207 cellular  17  nov      977         3    -1         0 unknown yes
## 45208 cellular  17  nov      456         2    -1         0 unknown yes
## 45209 cellular  17  nov     1127         5   184         3 success yes
## 45210 telephone 17  nov      508         4    -1         0 unknown no
## 45211 cellular  17  nov      361         2   188        11  other  no
```

```
str(bank_data, give.attr = FALSE)
```

```
## 'data.frame':   45211 obs. of  17 variables:
## $ age      : int  58 44 33 47 33 35 28 42 58 43 ...
## $ job      : chr   "management" "technician" "entrepreneur" "blue-collar" ...
## $ marital   : chr   "married" "single" "married" "married" ...
## $ education: chr   "tertiary" "secondary" "secondary" "unknown" ...
## $ default   : chr   "no" "no" "no" "no" ...
## $ balance   : int  2143 29 2 1506 1 231 447 2 121 593 ...
## $ housing   : chr   "yes" "yes" "yes" "yes" ...
## $ loan      : chr   "no" "no" "yes" "no" ...
## $ contact   : chr   "unknown" "unknown" "unknown" "unknown" ...
## $ day       : int   5 5 5 5 5 5 5 5 5 5 ...
## $ month     : chr   "may" "may" "may" "may" ...
## $ duration  : int  261 151 76 92 198 139 217 380 50 55 ...
## $ campaign  : int   1 1 1 1 1 1 1 1 1 1 ...
## $ pdays     : int  -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 ...
## $ previous  : int   0 0 0 0 0 0 0 0 0 0 ...
## $ poutcome  : chr   "unknown" "unknown" "unknown" "unknown" ...
```

```
## $ y      : chr  "no" "no" "no" "no" ...
```

A first look at the data shows us that many of the provided columns are marked with an incorrect data type. For example, default and marital status are set as character data types when they should be factors.

```
factor_cols <- c("marital", "job", "education", "contact", "poutcome", "day")
logical_cols <- c("default", "housing", "loan", "y")
months <- c("jan", "feb", "mar", "apr", "may", "jun",
            "jul", "aug", "sep", "oct", "nov", "dec")
```

```
bank_data <- bank_data %>%
  mutate(
    job = if_else(job == "admin.", "admin", job),
    across(all_of(factor_cols), as.factor),
    across(all_of(logical_cols), ~ .x == "yes"),
    month = factor(month, levels = months),
    job = relevel(job, ref = "unemployed"),
    marital = relevel(marital, ref = "single"),
    education = relevel(education, ref = "unknown"),
    contact = relevel(contact, ref = "unknown"),
    poutcome = relevel(poutcome, ref = "unknown")
  )
```

```
str(bank_data, give.attr = FALSE)
```

```
## 'data.frame': 45211 obs. of 17 variables:
## $ age      : int  58 44 33 47 33 35 28 42 58 43 ...
## $ job      : Factor w/ 12 levels "unemployed","admin",...: 6 11 4 3 12 6 6 4 7 11 ...
## $ marital  : Factor w/ 3 levels "single","divorced",...: 3 1 3 3 1 3 1 2 3 1 ...
## $ education: Factor w/ 4 levels "unknown","primary",...: 4 3 3 1 1 4 4 4 2 3 ...
## $ default  : logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
## $ balance  : int  2143 29 2 1506 1 231 447 2 121 593 ...
## $ housing  : logi  TRUE TRUE TRUE TRUE FALSE TRUE ...
## $ loan     : logi  FALSE FALSE TRUE FALSE FALSE FALSE ...
## $ contact  : Factor w/ 3 levels "unknown","cellular",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ day      : Factor w/ 31 levels "1","2","3","4",...: 5 5 5 5 5 5 5 5 5 5 ...
## $ month    : Factor w/ 12 levels "jan","feb","mar",...: 5 5 5 5 5 5 5 5 5 5 ...
## $ duration : int  261 151 76 92 198 139 217 380 50 55 ...
## $ campaign : int  1 1 1 1 1 1 1 1 1 1 ...
## $ pdays    : int  -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 ...
## $ previous : int  0 0 0 0 0 0 0 0 0 0 ...
## $ poutcome : Factor w/ 4 levels "unknown","failure",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ y        : logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
```

We have 45211 rows and 16 columns (excluding y).

A look at the description by the researchers tells us that there are no missing values even though some columns have values “unknown”. We must decide whether to keep them as “unknown” or convert them to NA. Either way, missing values must be inspected.

```
sum(rowSums(bank_data == "unknown") > 0)
```

```
## [1] 37369
```

There are a total of 37369 rows with at least one “unknown” value.

How many “unknowns” does each column have?

```
count_unknown(bank_data, "unknown")
```

```
## # A tibble: 4 x 2
##   column    count
##   <chr>    <int>
## 1 poutcome 36959
## 2 contact  13020
## 3 education 1857
## 4 job       288
```

Almost all poutcome column values are unknown. Let's keep this column for now as we will look at outcome distributions regarding y (subscription outcome) values later on.

Lastly, since some columns have names that may be difficult to interpret without looking at the metadata first, we should rename them.

```
bank_data <- bank_data %>%
  rename(in_default = "default",
         housing_loan = "housing",
         personal_loan = "loan",
         contact_type = "contact",
         subscribed = "y")
```

```
lapply(bank_data[, !(names(bank_data) %in% c("age", "balance", "duration", "pdays"))],
       unique)
```

```
## $job
## [1] management    technician    entrepreneur blue-collar  unknown
## [6] retired        admin          services      self-employed unemployed
## [11] housemaid      student
## 12 Levels: unemployed admin blue-collar entrepreneur housemaid ... unknown
##
## $marital
## [1] married single divorced
## Levels: single divorced married
##
## $education
## [1] tertiary secondary unknown primary
## Levels: unknown primary secondary tertiary
##
## $in_default
## [1] FALSE TRUE
##
## $housing_loan
## [1] TRUE FALSE
##
## $personal_loan
## [1] FALSE TRUE
##
## $contact_type
## [1] unknown cellular telephone
## Levels: unknown cellular telephone
##
## $day
## [1] 5 6 7 8 9 12 13 14 15 16 19 20 21 23 26 27 28 29 30 2 3 4 11 17 18
## [26] 24 25 1 10 22 31
```

```

## 31 Levels: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 ... 31
##
## $month
## [1] may jun jul aug oct nov dec jan feb mar apr sep
## Levels: jan feb mar apr may jun jul aug sep oct nov dec
##
## $campaign
## [1] 1 2 3 5 4 6 7 8 9 10 11 12 13 19 14 24 16 32 18 22 15 17 25 21 43
## [26] 51 63 41 26 28 55 50 38 23 20 29 31 37 30 46 27 58 33 35 34 36 39 44
##
## $previous
## [1] 0 3 1 4 2 11 16 6 5 10 12 7 18 9 21 8 14 15 26
## [20] 37 13 25 20 27 17 23 38 29 24 51 275 22 19 30 58 28 32 40
## [39] 55 35 41
##
## $poutcome
## [1] unknown failure other success
## Levels: unknown failure other success
##
## $subscribed
## [1] FALSE TRUE

```

Looking at the unique columns values we do not see anything out of the ordinary.

Exploratory analysis

Now we can investigate each variable separately.

Subscribed

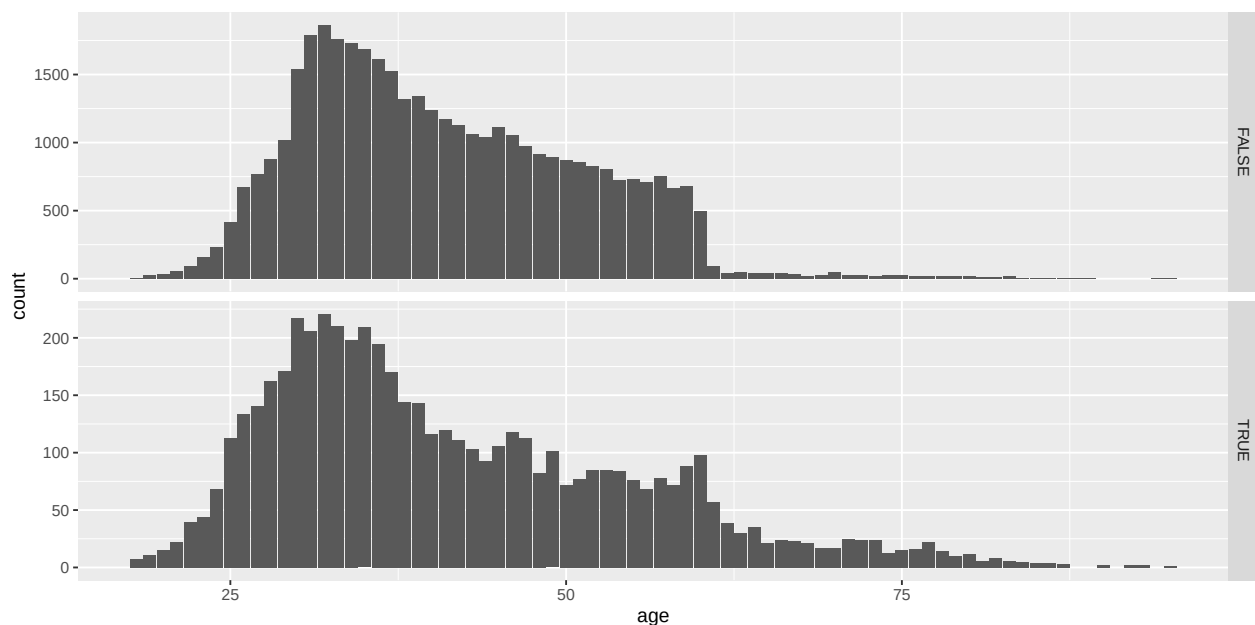
```
table(bank_data$subscribed)
```

```
##  
## FALSE  TRUE  
## 39922  5289
```

We have a severe imbalance in our data. Only 11,6% of contacted clients subscribed. We must also take this into account when removing unknown/NA values.

Age

```
ggplot(bank_data, aes(x = age)) +  
  geom_bar() +  
  facet_grid(subscribed ~ ., scales = "free_y")
```



Many clients contacted by the bank were between 25 and 60 years old. Age here is not distributed normally. Using these insights we can create a categorical age variable by segmenting then into young, working-age and older customers.

```
bank_data <- bank_data %>%  
  mutate(  
    age_categ = case_when(age > 60 ~ "high", age > 25 ~ "mid", TRUE ~ "low"),  
    age_categ = factor(age_categ),  
    age_categ = relevel(age_categ, ref = "low")  
  )  
  
CrossTable(bank_data$subscribed, bank_data$age_categ, prop.t = FALSE, prop.chisq = FALSE)
```

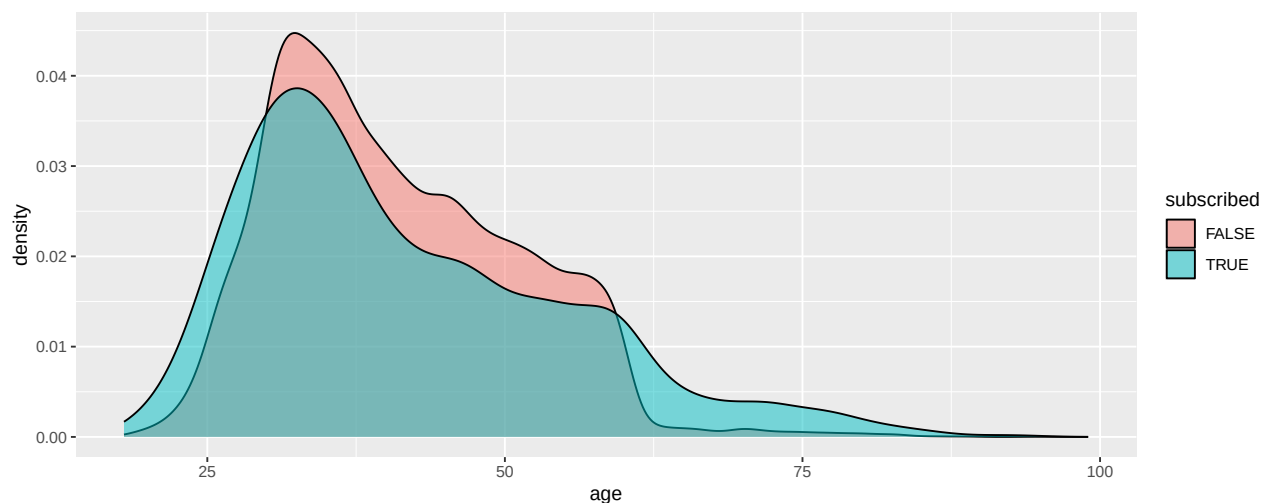
```
##
```

```
##
##      Cell Contents
## |-----|
## |                N |
## |      N / Row Total |
## |      N / Col Total |
## |-----|
##
##
## Total Observations in Table:  45211
##
##
##      | bank_data$age_categ
## bank_data$subscribed |      low |      high |      mid | Row Total |
## -----|-----|-----|-----|-----|
##                FALSE |      1016 |        686 |      38220 |      39922 |
##                |      0.025 |      0.017 |      0.957 |      0.883 |
##                |      0.760 |      0.577 |      0.895 |      |
## -----|-----|-----|-----|-----|
##                TRUE  |       320 |        502 |       4467 |       5289 |
##                |      0.061 |      0.095 |      0.845 |      0.117 |
##                |      0.240 |      0.423 |      0.105 |      |
## -----|-----|-----|-----|-----|
##      Column Total |      1336 |       1188 |      42687 |      45211 |
##                |      0.030 |      0.026 |      0.944 |      |
## -----|-----|-----|-----|-----|
##
##
```

Clients of at least the age of 60 were most likely to subscribe: 42,3% of them chose to do so. That is the highest percentage of all age groups even though older clients make up the smallest part of the total population.

The previous (continuous) age variable does not indicate a linear relationship between age and subscription rates.

```
ggplot(bank_data, aes(x = age, fill = subscribed)) +
  geom_density(alpha = 0.5) +
  xlim(18, 99)
```



The density plots also do not show a large difference in terms of age with the exception being clients over the age of 60.

Job

```
summary(bank_data$job)
```

```
##      unemployed      admin  blue-collar  entrepreneur      housemaid  
##          1303         5171         9732          1487         1240  
##      management      retired  self-employed      services      student  
##          9458         2264         1579          4154          938  
##      technician      unknown  
##          7597          288
```

There is a total of 228 unknown job values. Due to the large number of total rows, we can afford to drop the “unknowns”. The inclusion “unknown” job type would make the model less stable. Looking at the bigger picture, unknown values in this column should not exist at all as the bank should have this information about their clients.

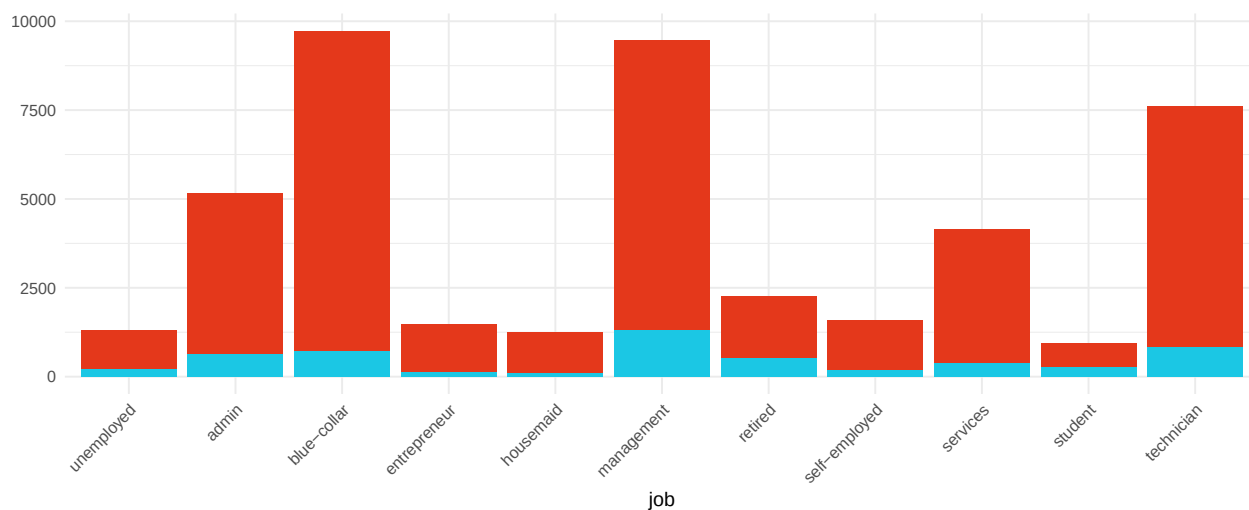
```
bank_data <- bank_data %>%  
  filter(job != "unknown") %>%  
  mutate(job = factor(job))
```

```
nrow(bank_data)
```

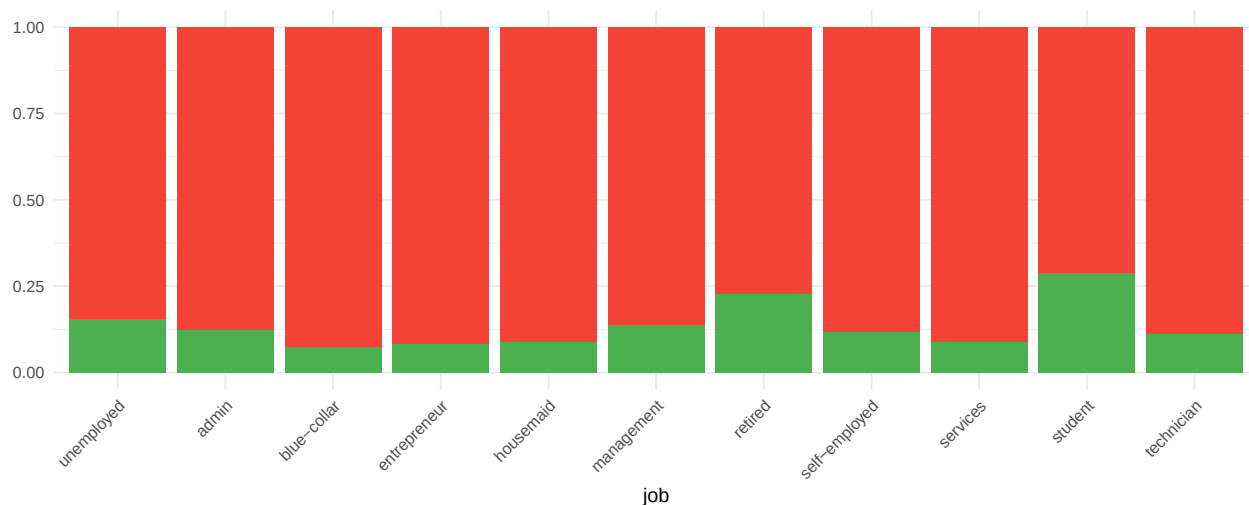
```
## [1] 44923
```

Let's look at what percentage of clients subscribed based on their job.

```
create_plot(bank_data, "job", "count")
```



```
create_plot(bank_data, "job")
```



As the chart shows, students, of all jobs, were most likely to subscribe to a deposit (28,7%) with retired workers following second at 22,8%. This could be to students having fewer major expenses, such as mortgages or car loans, and being heavily dependent on their parents. Retirees also often seek low-risk investment options, making bank deposits attractive.

```
CrossTable(bank_data$job, bank_data$subscribed, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
##
##      Cell Contents
## |-----|
## |              N |
## |              N / Row Total |
## |              N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44923
##
##
##      | bank_data$subscribed
## bank_data$job |      FALSE |      TRUE | Row Total |
## -----|-----|-----|-----|
##      unemployed |      1101 |       202 |      1303 |
##      |      0.845 |      0.155 |      0.029 |
##      |      0.028 |      0.038 |      |
## -----|-----|-----|
##      admin |      4540 |       631 |      5171 |
##      |      0.878 |      0.122 |      0.115 |
##      |      0.114 |      0.120 |      |
## -----|-----|-----|
##      blue-collar |      9024 |       708 |      9732 |
##      |      0.927 |      0.073 |      0.217 |
##      |      0.227 |      0.135 |      |
## -----|-----|-----|
##      entrepreneur |      1364 |       123 |      1487 |
##      |      0.917 |      0.083 |      0.033 |
##      |      0.034 |      0.023 |      |
```

##	-----	-----	-----	-----
##	housemaid	1131	109	1240
##		0.912	0.088	0.028
##		0.029	0.021	
##	-----	-----	-----	-----
##	management	8157	1301	9458
##		0.862	0.138	0.211
##		0.206	0.248	
##	-----	-----	-----	-----
##	retired	1748	516	2264
##		0.772	0.228	0.050
##		0.044	0.098	
##	-----	-----	-----	-----
##	self-employed	1392	187	1579
##		0.882	0.118	0.035
##		0.035	0.036	
##	-----	-----	-----	-----
##	services	3785	369	4154
##		0.911	0.089	0.092
##		0.095	0.070	
##	-----	-----	-----	-----
##	student	669	269	938
##		0.713	0.287	0.021
##		0.017	0.051	
##	-----	-----	-----	-----
##	technician	6757	840	7597
##		0.889	0.111	0.169
##		0.170	0.160	
##	-----	-----	-----	-----
##	Column Total	39668	5255	44923
##		0.883	0.117	
##	-----	-----	-----	-----
##				
##				

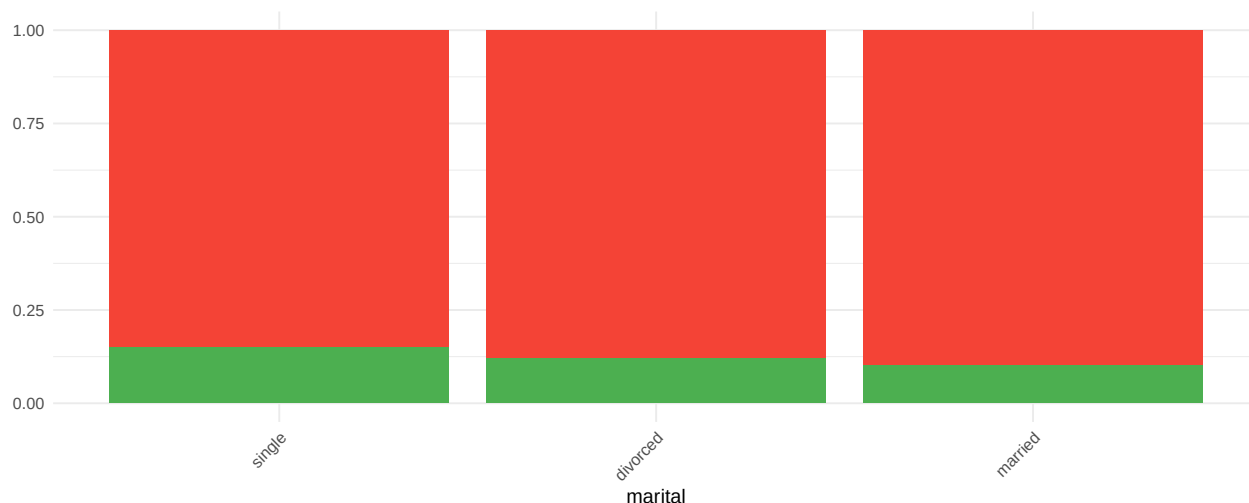
Marital status

```
CrossTable(bank_data$subscribed, bank_data$marital, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
##
##   Cell Contents
## |-----|
## |                N |
## |      N / Row Total |
## |      N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44923
##
##
##               | bank_data$marital
## bank_data$subscribed |   single |   divorced |   married | Row Total |
## -----|-----|-----|-----|-----|
##             FALSE |   10822 |     4569 |    24277 |   39668 |
##             |   0.273 |     0.115 |     0.612 |     0.883 |
##             |   0.851 |     0.880 |     0.899 |           |
## -----|-----|-----|-----|-----|
##             TRUE  |    1900 |      621 |     2734 |    5255 |
##             |   0.362 |     0.118 |     0.520 |     0.117 |
##             |   0.149 |     0.120 |     0.101 |           |
## -----|-----|-----|-----|-----|
##      Column Total |   12722 |     5190 |    27011 |   44923 |
##             |   0.283 |     0.116 |     0.601 |           |
## -----|-----|-----|-----|-----|
##
##
##
```

Married clients make up 60,1% of our data set. Single clients were slightly more likely to subscribe to a deposit (14,9%) than other clients. It is also probable that this tendency is caused by randomness as marital status categories are not divided equally (single - 28,3%, divorced - 11,6% and married - 60,1%).

```
create_plot(bank_data, "marital")
```



Education

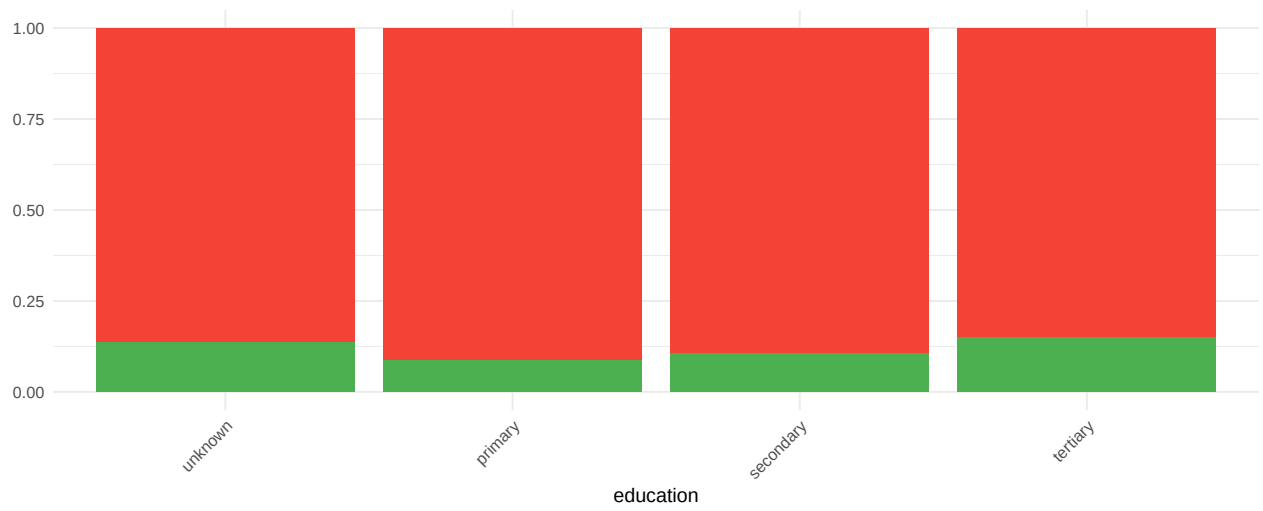
```
CrossTable(bank_data$subscribed, bank_data$education, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
##
##   Cell Contents
## |-----|
## |                N |
## |      N / Row Total |
## |      N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44923
##
##
##               | bank_data$education
## bank_data$subscribed |   unknown |   primary | secondary | tertiary | Row Total |
## -----|-----|-----|-----|-----|-----|
##                FALSE |    1496 |    6212 |    20690 |    11270 |    39668 |
##                |    0.038 |    0.157 |    0.522 |    0.284 |    0.883 |
##                |    0.865 |    0.914 |    0.894 |    0.850 |           |
## -----|-----|-----|-----|-----|-----|
##                TRUE  |     234 |     588 |     2441 |     1992 |     5255 |
##                |    0.045 |    0.112 |    0.465 |    0.379 |    0.117 |
##                |    0.135 |    0.086 |    0.106 |    0.150 |           |
## -----|-----|-----|-----|-----|-----|
##      Column Total |    1730 |    6800 |    23131 |    13262 |    44923 |
##                |    0.039 |    0.151 |    0.515 |    0.295 |           |
## -----|-----|-----|-----|-----|-----|
##
##
```

There are 1730 “unknown” values (3,9%) in the education variable. If we removed these “unknowns” we would risk causing further imbalance in the subscribed variable as only 5255 (11,7%) of clients decided to make a deposit subscription in total (234 of them had an “unknown” education).

Clients with a tertiary (college/university/vocational training) education (29,5%) are most likely to subscribe out of all groups - 15% of them chose to do so.

```
create_plot(bank_data, "education")
```



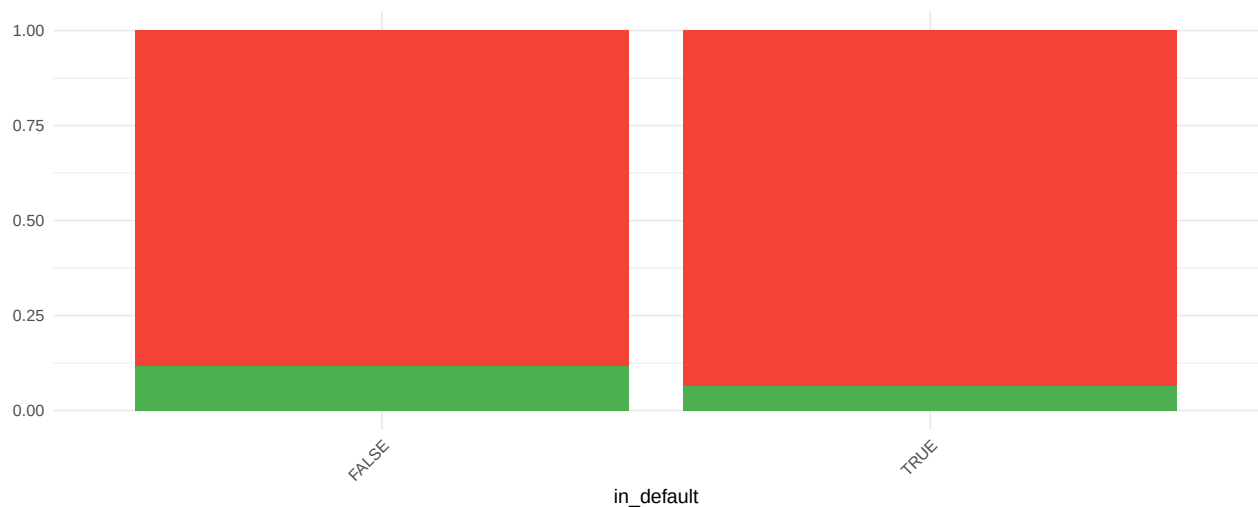
Default status

```
CrossTable(bank_data$subscribed, bank_data$in_default, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
##
##   Cell Contents
## |-----|
## |                N |
## |      N / Row Total |
## |      N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44923
##
##
##               | bank_data$in_default
## bank_data$subscribed |      FALSE |      TRUE | Row Total |
## -----|-----|-----|-----|
##                FALSE |      38907 |       761 |      39668 |
##                |      0.981 |      0.019 |      0.883 |
##                |      0.882 |      0.936 |            |
## -----|-----|-----|-----|
##                TRUE |       5203 |        52 |       5255 |
##                |      0.990 |      0.010 |      0.117 |
##                |      0.118 |      0.064 |            |
## -----|-----|-----|-----|
##      Column Total |      44110 |        813 |      44923 |
##                |      0.982 |      0.018 |            |
## -----|-----|-----|-----|
##
##
```

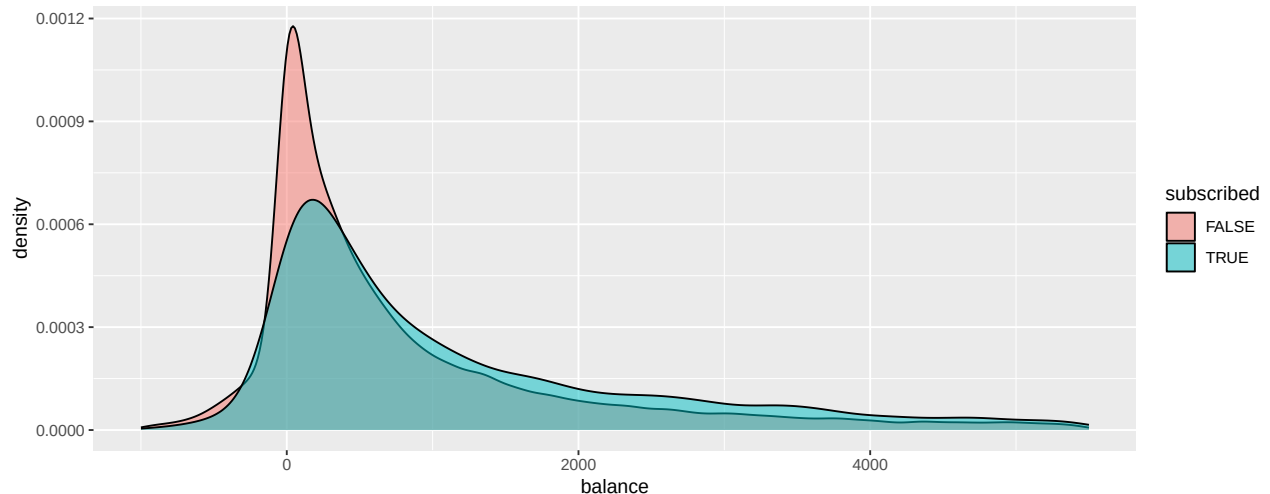
Only 6,4% of clients that were in default chose to sign up for a deposit. Out of the total sample only 1,8% of clients were in default. This variable is likely to be a good indicator of whether the client subscribes to a deposit, but we most likely should not offer deposits to clients that are in default at all.

```
create_plot(bank_data, "in_default")
```



Balance (yearly average)

```
ggplot(bank_data, aes(x = balance, fill = subscribed)) +  
  geom_density(alpha = 0.5) +  
  xlim(-1000, 5500)
```



The balance density plot does not immediately indicate that wealthier clients are more likely to make a subscription. We can try to create a new indicator variable that checks if the client has a balance above the peak balance for subscribed clients.

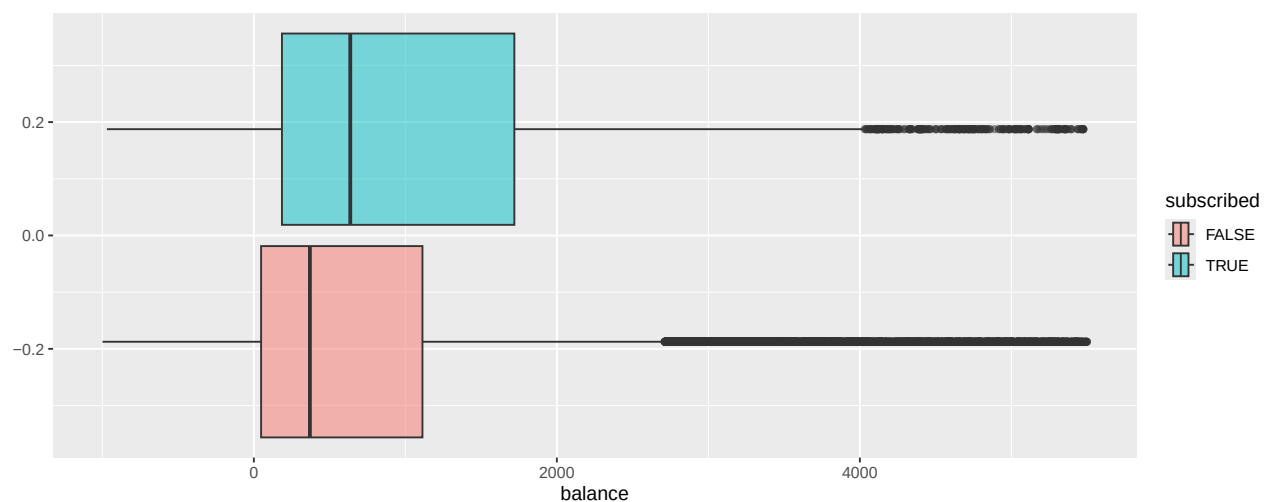
```
density <- density(bank_data$balance[bank_data$subscribed == TRUE])  
density$x[which.max(density$y)]
```

```
## [1] 258.2939
```

The distribution peaks at around 258 euros. We'll pick 250.

```
bank_data <- bank_data %>%  
  mutate(balance_above_250 = balance > 250)
```

```
ggplot(bank_data, aes(x = balance, fill = subscribed)) +  
  geom_boxplot(alpha = 0.5) +  
  xlim(-1000, 5500)
```



Since we are dealing with financial data, there are many extreme outliers in the distributions of variables.

Even though the box plots do not paint the full picture, they do indicate that the median balance is higher for those who chose to subscribe.

```
paste0("Balance Mean: ", round(mean(bank_data$balance, na.rm = TRUE), 2))
```

```
## [1] "Balance Mean: 1359.64"
```

```
paste0("Balance Standard Deviation: ", round(sd(bank_data$balance), 2))
```

```
## [1] "Balance Standard Deviation: 3045.09"
```

```
outliers <- boxplot.stats(bank_data$balance)$out
```

```
outlierNum <- length(outliers)
```

```
paste0("Outlier Percentage: ", round(outlierNum/(length(bank_data$balance)) * 100, 2))
```

```
## [1] "Outlier Percentage: 10.49"
```

Since the balance standard deviation is relatively high (3045,09 euros) and 10,49% of the entries can be marked as outliers, we'll normalize the balance variable using the Order-Norm transformation (converts each value to its percentile rank in the original distribution, then maps that percentile to the corresponding value in a standard normal distribution). This will provide at least a rough comparison between the groups.

```
on <- orderNorm(bank_data$balance)
```

```
bank_data$trans_balance <- predict(on)
```

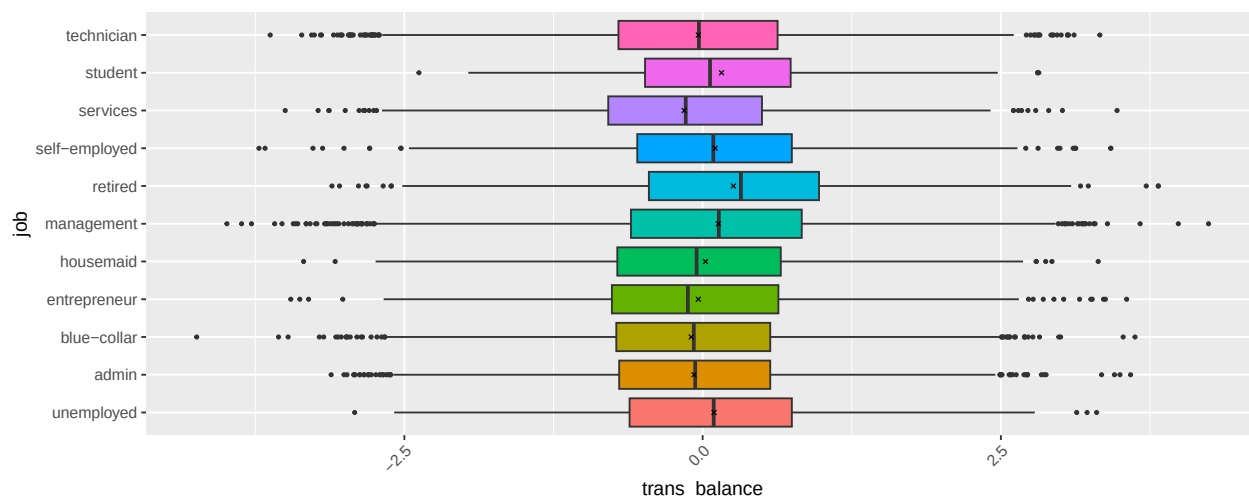
```
ggplot(bank_data, aes(x = job, y = trans_balance, fill = job)) +
```

```
  geom_boxplot(outlier.size = 0.7, na.rm = TRUE) +
```

```
  coord_flip() +
```

```
  stat_summary(fun = mean, geom = "point", shape = 4, size = 0.8, color = "black", na.rm = TRUE) +
```

```
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none")
```



The box plots allow us to conclude that the balance of client accounts is likely dependent more factors than simply their job. It also indicates that the clients, grouped by their job type, are not homogeneous (even with balance transformed). Nevertheless, we can draw certain conclusions. For example, we can see that the retirees have the highest average and median balance and service workers have the lowest balance. This is logical as retirees often have accumulated savings over their lifetime, while service workers may have lower wages and less opportunity to save.

Housing and Personal loans

```
CrossTable(bank_data$subscribed, bank_data$housing_loan, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
##
##   Cell Contents
## |-----|
## |                N |
## |      N / Row Total |
## |      N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44923
##
##
##               | bank_data$housing_loan
## bank_data$subscribed |    FALSE |    TRUE | Row Total |
## -----|-----|-----|-----|
##                FALSE |    16497 |    23171 |    39668 |
##                |    0.416 |    0.584 |    0.883 |
##                |    0.832 |    0.923 |           |
## -----|-----|-----|-----|
##                TRUE |     3322 |     1933 |     5255 |
##                |    0.632 |    0.368 |    0.117 |
##                |    0.168 |    0.077 |           |
## -----|-----|-----|-----|
##      Column Total |    19819 |    25104 |    44923 |
##                |    0.441 |    0.559 |           |
## -----|-----|-----|-----|
##
##
```

55,9% of the clients in our sample had a housing loan. Clients that did not have a housing loan were more than twice as likely to subscribe than the clients without one. This variable will be significant when modelling.

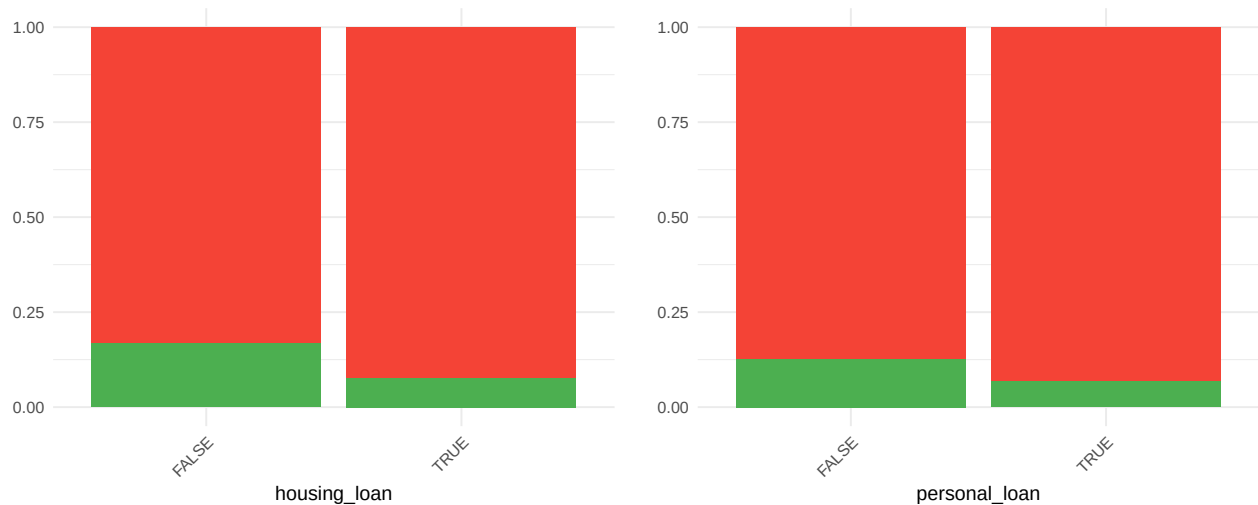
```
CrossTable(bank_data$subscribed, bank_data$personal_loan, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
##
##   Cell Contents
## |-----|
## |                N |
## |      N / Row Total |
## |      N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44923
##
##
##               | bank_data$personal_loan
## bank_data$subscribed |    FALSE |    TRUE | Row Total |
## -----|-----|-----|-----|
```

```
##          FALSE |      32910 |      6758 |      39668 |
##          |      0.830 |      0.170 |      0.883 |
##          |      0.873 |      0.933 |          |
## -----|-----|-----|-----|
##          TRUE |      4773 |      482 |      5255 |
##          |      0.908 |      0.092 |      0.117 |
##          |      0.127 |      0.067 |          |
## -----|-----|-----|-----|
##      Column Total |      37683 |      7240 |      44923 |
##          |      0.839 |      0.161 |          |
## -----|-----|-----|-----|
##
##
```

The situation with personal loans is practically the same as with housing loans accept the fact that only 16,1% of the clients had a personal loan (55,9% had a housing loan). Clients that did not have a personal loan were 1,9 times as likely to subscribe than the clients without one.

```
plot_list <- lapply(c("housing_loan", "personal_loan"), function(var) create_plot(bank_data, var))
bar_plot_matrix <- grid.arrange(grobs = plot_list, ncol = 2)
```



It is clear that this variable will also be significant when modelling as clients with no financial burdens (defaults and loans) are more likely to subscribe.

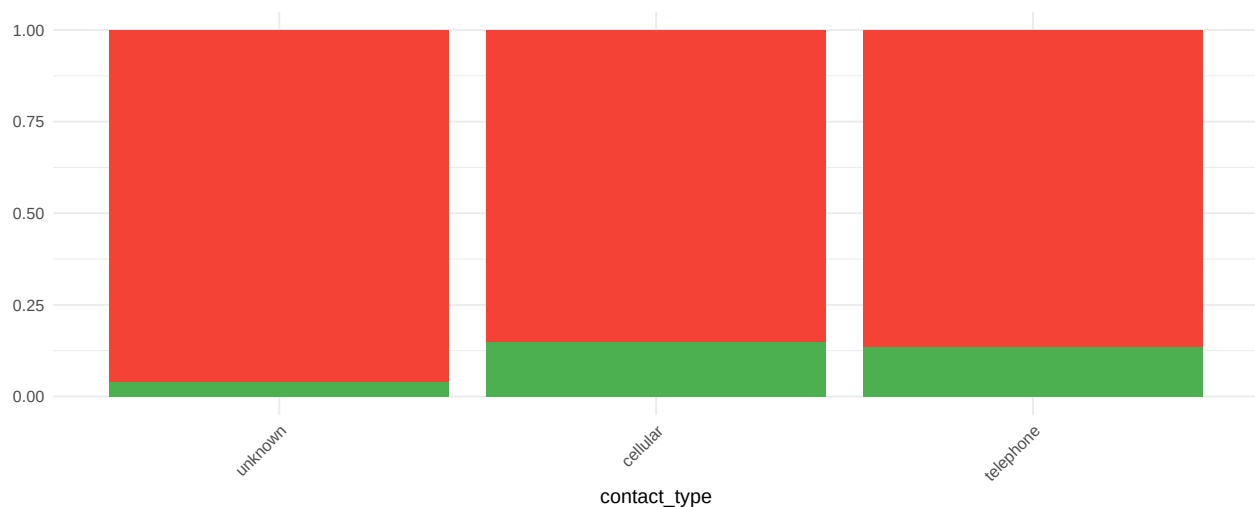
Contact type

```
CrossTable(bank_data$subscribed, bank_data$contact_type, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
##
##   Cell Contents
## |-----|
## |               N |
## |       N / Row Total |
## |       N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44923
##
##
##               | bank_data$contact_type
## bank_data$subscribed |   unknown |   cellular |   telephone | Row Total |
## -----|-----|-----|-----|-----|
##               FALSE |    12381 |    24812 |     2475 |    39668 |
##               |     0.312 |     0.625 |     0.062 |     0.883 |
##               |     0.959 |     0.851 |     0.865 |           |
## -----|-----|-----|-----|-----|
##               TRUE  |     528 |     4342 |       385 |     5255 |
##               |     0.100 |     0.826 |     0.073 |     0.117 |
##               |     0.041 |     0.149 |     0.135 |           |
## -----|-----|-----|-----|-----|
##           Column Total |    12909 |    29154 |     2860 |    44923 |
##               |     0.287 |     0.649 |     0.064 |           |
## -----|-----|-----|-----|-----|
##
##
```

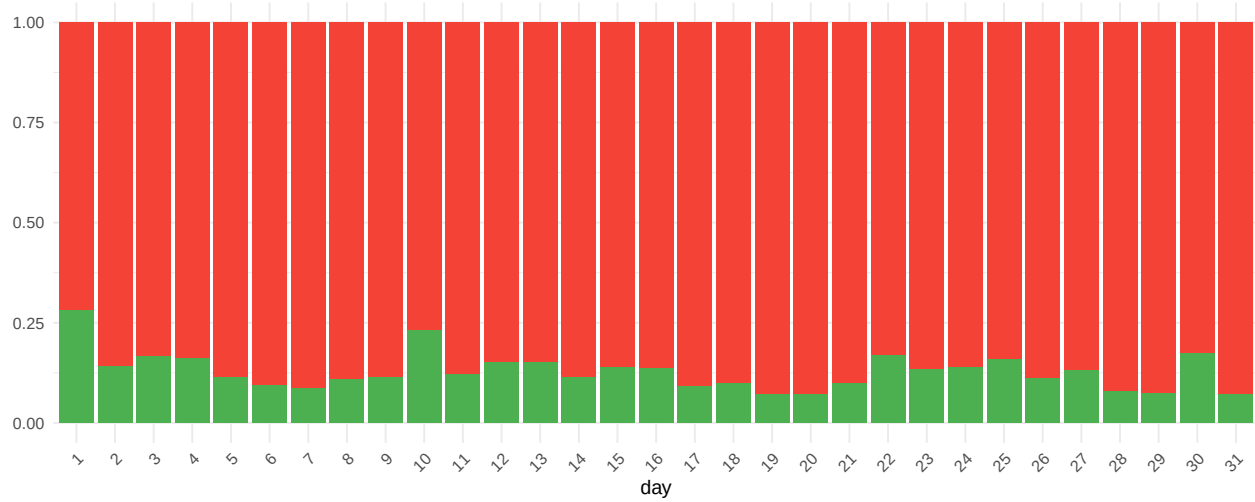
Clients that were contacted through cellular were slightly more likely to make a subscription. The contact type for 28,7% of the clients is unknown. “Unknown” values here are an error in the data collection process. Anyway, this variable will not be useful as we would learn about the contact type only after contact.

```
create_plot(bank_data, "contact_type")
```

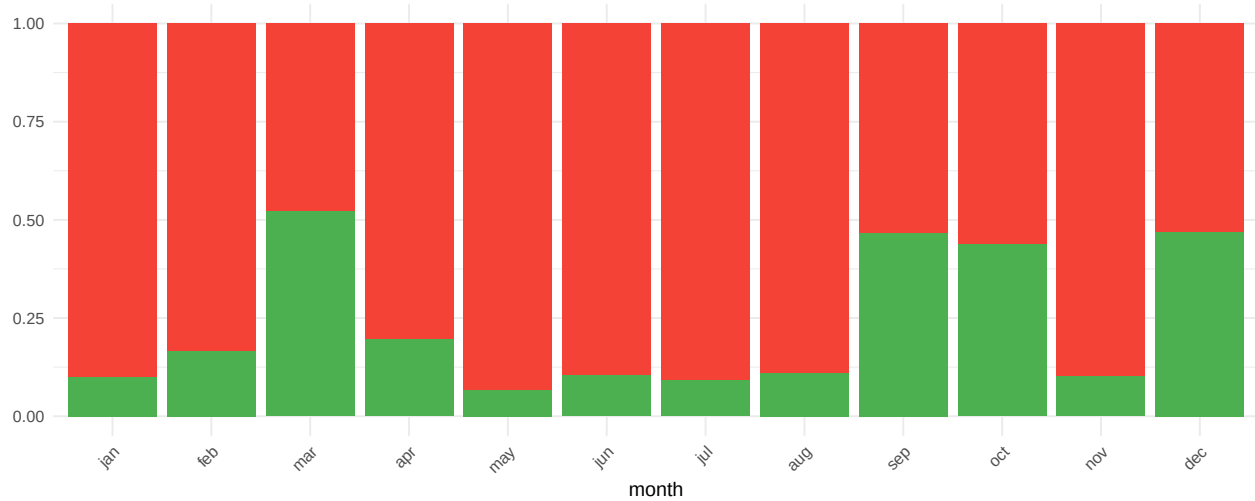


Day and month

```
create_plot(bank_data, "day")
```

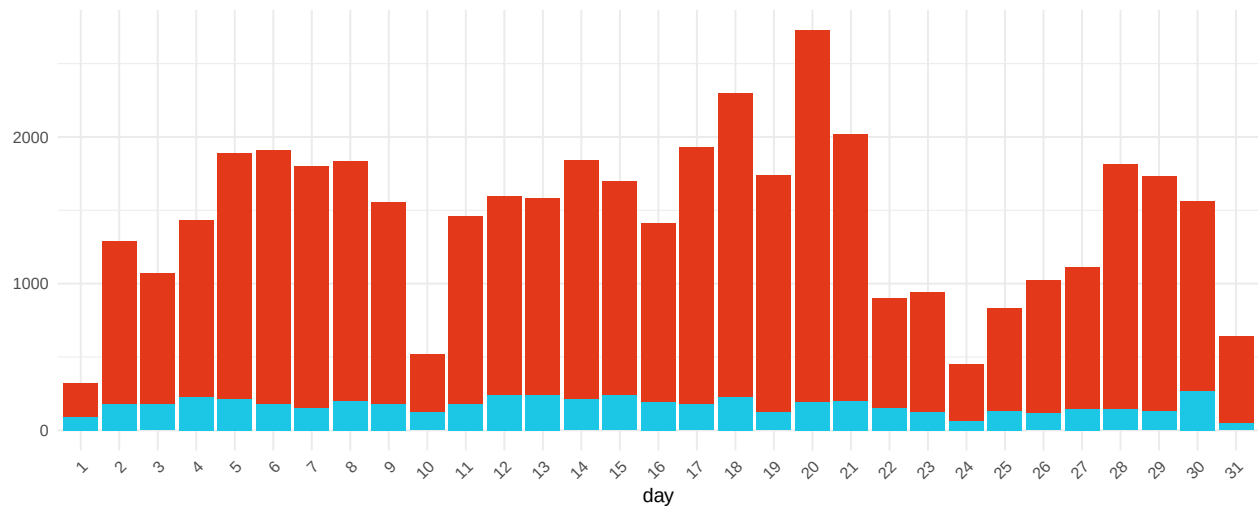


```
create_plot(bank_data, "month")
```

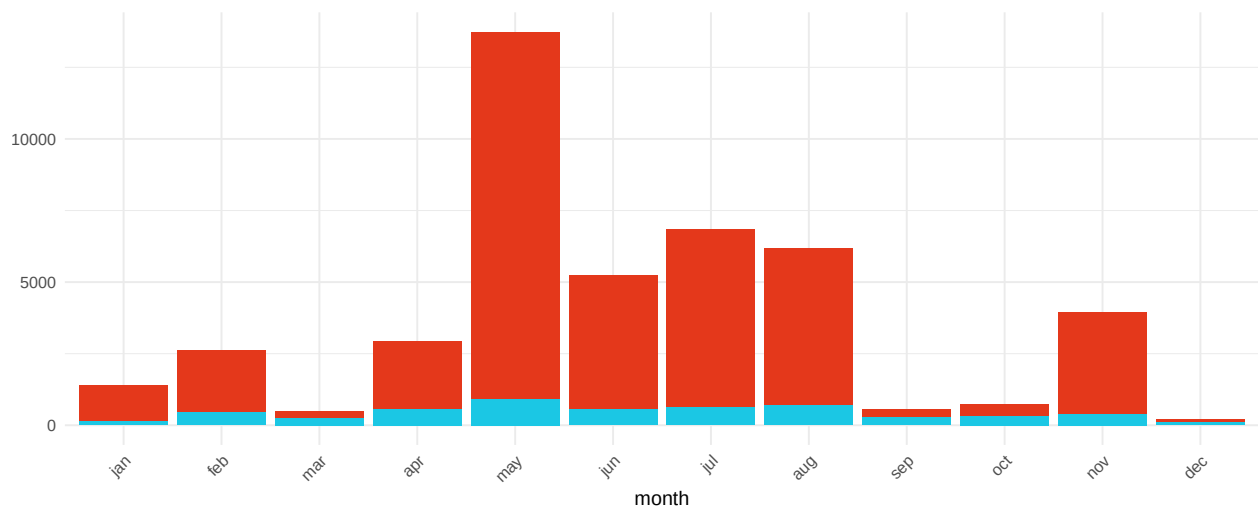


At first sight March, September, October and December seem to be the best months to contact the clients. Higher success could also be achieved when contacting the clients on the 1st, 10th, 22nd and 30th.

```
create_plot(bank_data, "day", "count")
```



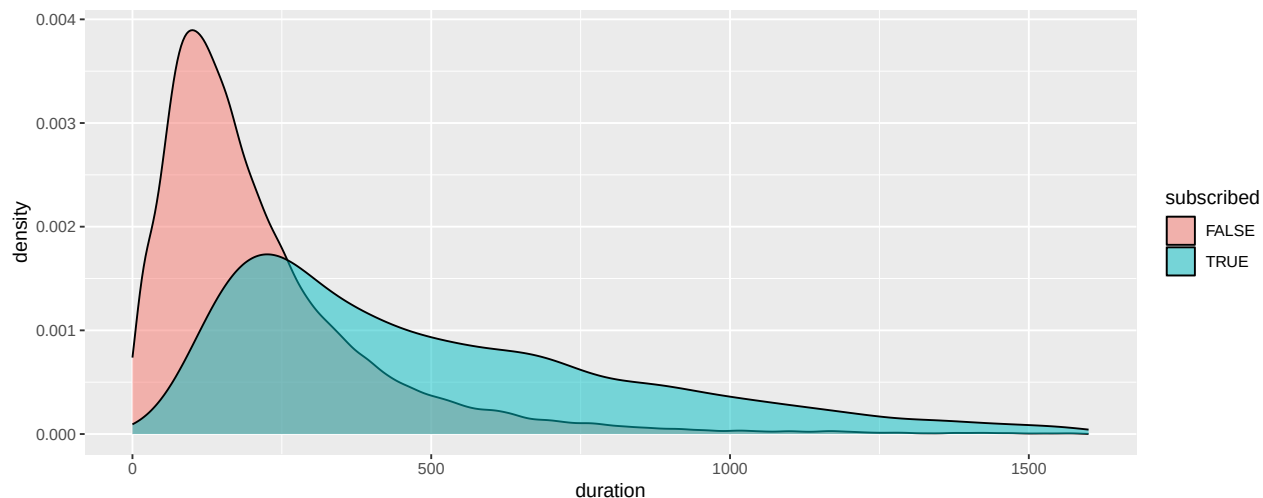
```
create_plot(bank_data, "month", "count")
```



After digging deeper, it becomes clear that the distribution of number of calls by date and month is disproportionate. The least clients were called on the 1st, 10th, 24th and 31st, and in March, September, October and December. This inconsistency should be addressed by the researchers that collected the data: more calls should be conducted to make the distribution more uniform.

Duration

```
ggplot(bank_data, aes(x = duration, fill = subscribed)) +  
  geom_density(alpha = 0.5) +  
  xlim(0, 1600)
```



Call duration seems to tell a clearer story than other continuous variables. Clients that, in the end, decided not to subscribe had shorter conversations with the representative of the bank showing their disinterest early on. Yet, we will not be able to use this variable as it appears only after a call has taken place (we are trying to pick which clients to call).

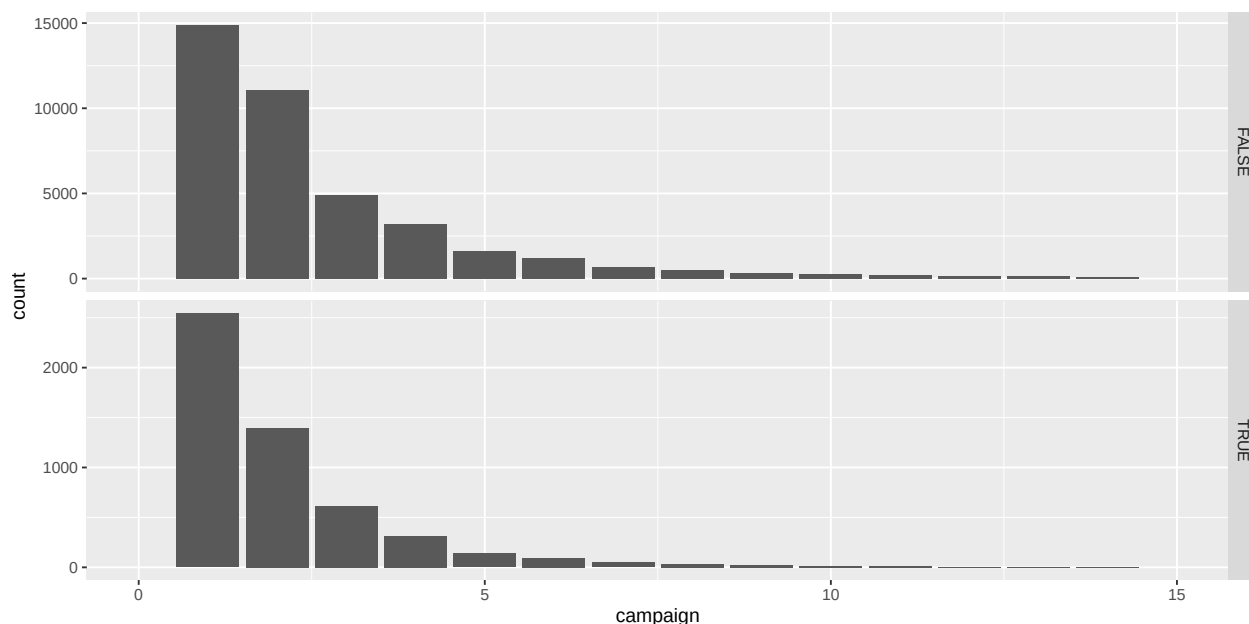
Attributes related to previous contact

Campaign contacts

```
ggplot(bank_data, aes(x = campaign)) +  
  geom_bar() +  
  facet_grid(subscribed ~ ., scales = "free_y") +  
  xlim(0, 15)
```

```
## Warning: Removed 525 rows containing non-finite outside the scale range  
## (`stat_count()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## (`geom_bar()`).
```



Warnings were kept on purpose, facet_grid does not knit properly without them.

Number of contacts performed during this campaign seems to be proportional to the number of contacts performed in total.

Let's look at how the number of total contacts is related to a successful deposit subscription.

```
subscribed_camp <- bank_data$subscribed[bank_data$campaign < 6]  
campaign_camp <- bank_data$campaign[bank_data$campaign < 6]  
  
CrossTable(subscribed_camp, campaign_camp, prop.t = FALSE, prop.chisq = FALSE)
```

```
##  
##  
##      Cell Contents  
## |-----|  
## |                      N |  
## |      N / Row Total |  
## |      N / Col Total |  
## |-----|  
##  
##
```

```
## Total Observations in Table: 40612
```

```
##
```

```
##
```

```
##          | campaign_camp
```

## subscribed_camp	1	2	3	4	5	Row Total
## -----	-----	-----	-----	-----	-----	-----
## FALSE	14896	11043	4873	3187	1610	35609
##	0.418	0.310	0.137	0.089	0.045	0.877
##	0.854	0.888	0.888	0.910	0.921	
## -----	-----	-----	-----	-----	-----	-----
## TRUE	2541	1395	613	315	139	5003
##	0.508	0.279	0.123	0.063	0.028	0.123
##	0.146	0.112	0.112	0.090	0.079	
## -----	-----	-----	-----	-----	-----	-----
## Column Total	17437	12438	5486	3502	1749	40612
##	0.429	0.306	0.135	0.086	0.043	
## -----	-----	-----	-----	-----	-----	-----

```
##
```

```
##
```

Number of contacts during the campaign seems to increase the likeliness of subscription but with diminishing returns.

Previous days

```
sum(bank_data$pdays != -1)
```

```
## [1] 8224
```

There are 8224 clients which have been contacted in the past. Since there are many different pdays (days passed after previous contact from last campaign) values and because the variable has been encoded as -1 or any other natural number, we can transform this variable into a binary variable.

```
bank_data <- bank_data %>%  
  mutate(was_contacted = ifelse(pdays == -1, FALSE, TRUE))
```

Previous contacts

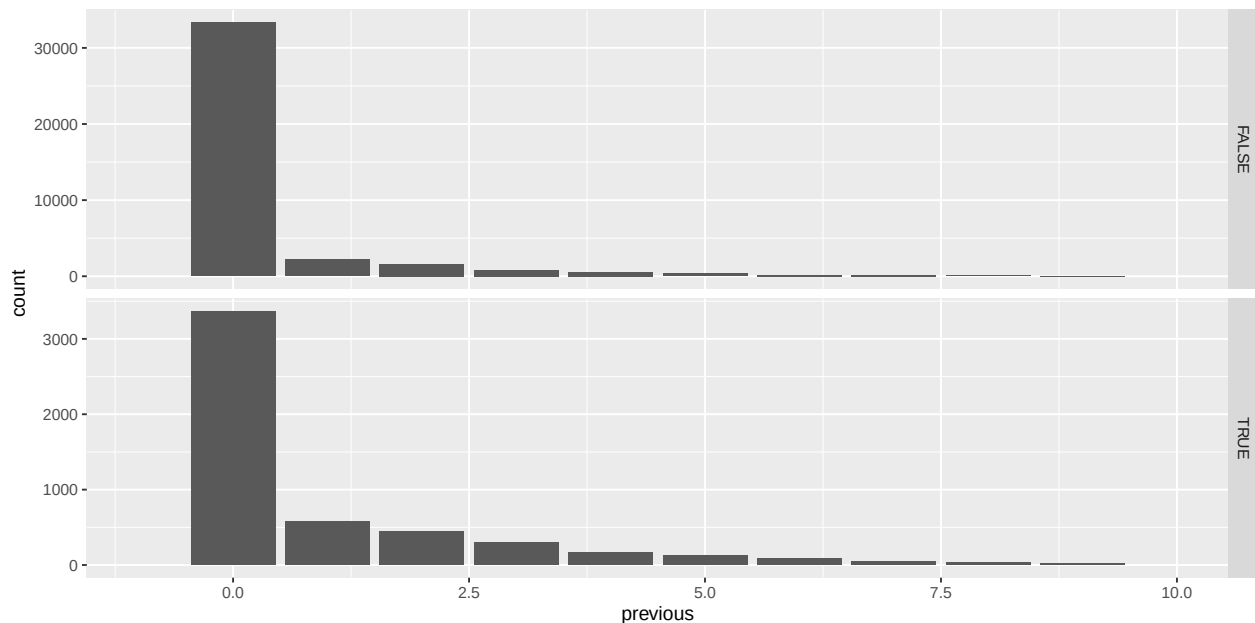
```
ggplot(bank_data, aes(x = previous)) +  
  geom_bar() +  
  facet_grid(subscribed ~ ., scales = "free_y") +  
  xlim(-1, 10)
```

```
## Warning: Removed 294 rows containing non-finite outside the scale range
```

```
## (`stat_count()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
```

```
## (`geom_bar()`).
```



```
# Warnings were kept on purpose, facet_grid does not knit properly without them.
```

```
subscribed_prev <- bank_data$subscribed[bank_data$previous < 7]  
previous_prev <- bank_data$previous[bank_data$previous < 7]
```

```
CrossTable(previous_prev, subscribed_prev, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
```

```
##
```

```
##      Cell Contents
```

```

## |-----|
## |                N |
## |      N / Row Total |
## |      N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44138
##
##
##          | subscribed_prev
## previous_prev |      FALSE |      TRUE | Row Total |
## -----|-----|-----|-----|
##          0 |    33333 |    3366 |    36699 |
##          |    0.908 |    0.092 |    0.831 |
##          |    0.853 |    0.665 |          |
## -----|-----|-----|-----|
##          1 |    2184 |    578 |    2762 |
##          |    0.791 |    0.209 |    0.063 |
##          |    0.056 |    0.114 |          |
## -----|-----|-----|-----|
##          2 |    1645 |    451 |    2096 |
##          |    0.785 |    0.215 |    0.047 |
##          |    0.042 |    0.089 |          |
## -----|-----|-----|-----|
##          3 |    847 |    292 |    1139 |
##          |    0.744 |    0.256 |    0.026 |
##          |    0.022 |    0.058 |          |
## -----|-----|-----|-----|
##          4 |    541 |    170 |    711 |
##          |    0.761 |    0.239 |    0.016 |
##          |    0.014 |    0.034 |          |
## -----|-----|-----|-----|
##          5 |    336 |    120 |    456 |
##          |    0.737 |    0.263 |    0.010 |
##          |    0.009 |    0.024 |          |
## -----|-----|-----|-----|
##          6 |    193 |    82 |    275 |
##          |    0.702 |    0.298 |    0.006 |
##          |    0.005 |    0.016 |          |
## -----|-----|-----|-----|
## Column Total |    39079 |    5059 |    44138 |
##          |    0.885 |    0.115 |          |
## -----|-----|-----|-----|
##
##

```

Number of contacts during the previous campaign seems to linearly increase the likeliness of subscription. This is due to the client being interested in subscribing to a deposit.

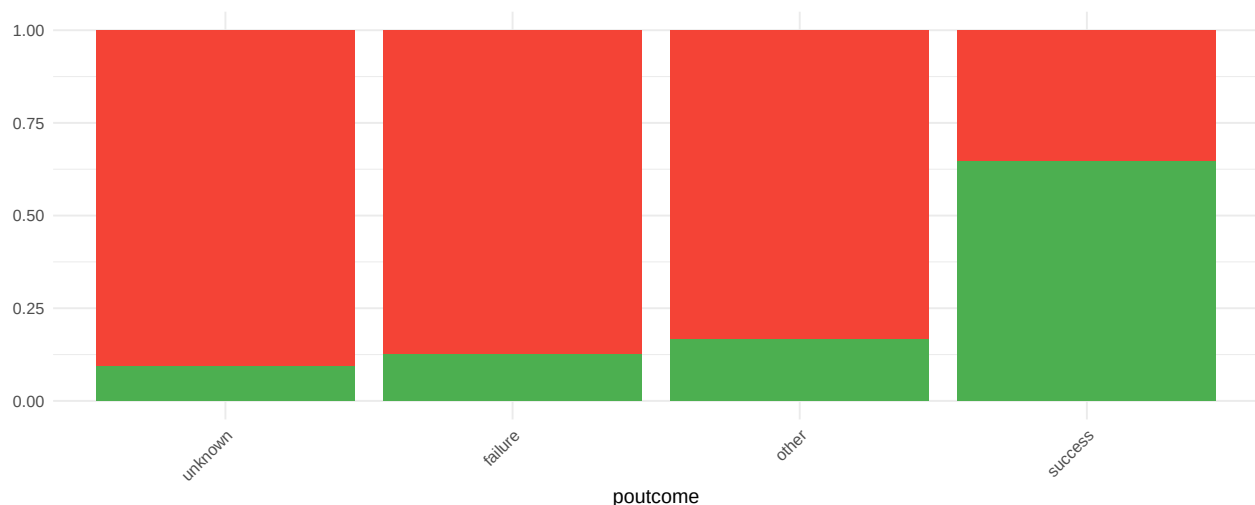
Previous outcome

```
CrossTable(bank_data$subscribed, bank_data$poutcome, prop.t = FALSE, prop.chisq = FALSE)
```

```
##
##
##   Cell Contents
## |-----|
## |                N |
## |      N / Row Total |
## |      N / Col Total |
## |-----|
##
##
## Total Observations in Table:  44923
##
##
##               | bank_data$poutcome
## bank_data$subscribed |   unknown |   failure |   other |   success | Row Total |
## -----|-----|-----|-----|-----|-----|
##               FALSE |    33336 |     4269 |     1532 |        531 |    39668 |
##               |    0.840 |     0.108 |     0.039 |     0.013 |     0.883 |
##               |    0.908 |     0.875 |     0.834 |     0.354 |           |
## -----|-----|-----|-----|-----|-----|
##               TRUE  |     3368 |        612 |        306 |        969 |     5255 |
##               |    0.641 |     0.116 |     0.058 |     0.184 |     0.117 |
##               |    0.092 |     0.125 |     0.166 |     0.646 |           |
## -----|-----|-----|-----|-----|-----|
##               Column Total |    36704 |     4881 |     1838 |        1500 |    44923 |
##               |    0.817 |     0.109 |     0.041 |     0.033 |           |
## -----|-----|-----|-----|-----|-----|
##
##
```

If the outcome of the previous campaign was successful, the outcome of the current campaign on the same client has a 64,6% likelihood of being successful. Although it must be noted that there are only 1500 clients with the poutcome attribute set as successful and the vast majority (81,7%) are set as unknown.

```
create_plot(bank_data, "poutcome")
```



Correlation of continuous variables

```
model <- lm(subscribed ~ age + balance + campaign + pdays + previous, data = bank_data)
vif_values <- car::vif(model)

print(vif_values)
```

```
##      age  balance campaign   pdays previous
## 1.010418 1.010211 1.008161 1.269709 1.260796
```

All VIF values are well below 4, indicating that there is no multicollinearity.

Modelling

A seed is set inside the `train_evaluate_model` function. Train-test data split of 80:20 also happens inside the function.

We will leave out `day`, `month`, `duration` and `contact_type` variables as they are either not useful or not available before contacting the client.

Let's select model parameters with backward stepwise selection and begin with variables that are less significant based on logic.

```
results_0 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con

## Setting levels: control = FALSE, case = TRUE

## Setting direction: controls < cases

cat("Model 0 (Full) Summary:\n")

## Model 0 (Full) Summary:

print(summary(results_0$model))

##
## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -6.496e-01  1.778e-01  -3.654 0.000258 ***
## age           -7.730e-03  2.368e-03  -3.264 0.001097 **
## jobadmin       -1.034e-02  1.088e-01  -0.095 0.924272
## jobblue-collar -2.576e-01  1.073e-01  -2.400 0.016381 *
## jobentrepreneur -2.528e-01  1.462e-01  -1.729 0.083731 .
## jobhousemaid   -4.044e-01  1.536e-01  -2.633 0.008460 **
## jobmanagement -7.337e-02  1.062e-01  -0.691 0.489745
## jobretired     -3.262e-02  1.320e-01  -0.247 0.804885
## jobself-employed -1.447e-01  1.344e-01  -1.077 0.281485
## jobservices    -1.987e-01  1.161e-01  -1.712 0.086848 .
## jobstudent      1.541e-01  1.392e-01   1.107 0.268345
## jobtechnician  -1.478e-01  1.053e-01  -1.403 0.160586
## maritaldivorced -5.500e-02  6.551e-02  -0.840 0.401151
## maritalmarried -2.697e-01  4.491e-02  -6.006 1.90e-09 ***
## educationprimary -9.416e-02  1.042e-01  -0.904 0.366094
## educationsecondary 3.428e-02  9.257e-02   0.370 0.711171
## educationtertiary 2.200e-01  9.708e-02   2.266 0.023461 *
## in_defaultTRUE    1.808e-01  1.584e-01   1.141 0.253673
## balance         -2.436e-05  9.047e-06  -2.692 0.007093 **
## housing_loanTRUE  -6.499e-01  3.853e-02 -16.867 < 2e-16 ***
## personal_loanTRUE -3.577e-01  5.797e-02  -6.169 6.86e-10 ***
## campaign         -8.761e-02  9.151e-03  -9.574 < 2e-16 ***
## pdays           3.131e-04  3.028e-04   1.034 0.301096
## previous         8.897e-03  6.106e-03   1.457 0.145061
## poutcomefailure -1.438e+00  1.009e+00  -1.425 0.154067
## poutcomeother   -1.147e+00  1.010e+00  -1.136 0.256064
## poutcomesuccess  7.247e-01  1.010e+00   0.718 0.473002
## age_catighigh    8.140e-01  1.504e-01   5.413 6.20e-08 ***
## age_catemid     -6.406e-01  9.326e-02  -6.869 6.49e-12 ***
```

```
## balance_above_250TRUE 7.659e-02 6.708e-02 1.142 0.253565
## trans_balance 2.136e-01 4.409e-02 4.844 1.27e-06 ***
## was_contactedTRUE 1.674e+00 1.011e+00 1.655 0.097861 .
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 25938 on 35938 degrees of freedom
## Residual deviance: 22414 on 35907 degrees of freedom
## AIC: 22478
##
## Number of Fisher Scoring iterations: 5
```

```
cat("\nModel 0 AUC:", round(results_0$auc, 4), "\n")
```

```
##
## Model 0 AUC: 0.7528
```

In the current model there are multiple versions of the age, the balance and the previous contact count variable. Let's try keeping only the best one out of each.

```
results_1 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
cat("Model 0 (Full) Summary:\n")
```

```
## Model 0 (Full) Summary:
```

```
print(summary(results_1$model))
```

```
##
## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.8562756  0.1730072  -4.949 7.45e-07 ***
## age         -0.0073125  0.0023626  -3.095  0.00197 **
## jobadmin     -0.0122179  0.1088442  -0.112  0.91062
## jobblue-collar -0.2590102  0.1073413  -2.413  0.01582 *
## jobentrepreneur -0.2518025  0.1461392  -1.723  0.08488 .
## jobhousemaid  -0.4003110  0.1536387  -2.606  0.00917 **
## jobmanagement -0.0700745  0.1062686  -0.659  0.50963
## jobretired    -0.0358000  0.1320658  -0.271  0.78633
## jobself-employed -0.1402946  0.1344106  -1.044  0.29659
## jobservices   -0.2016815  0.1161005  -1.737  0.08236 .
## jobstudent     0.1672463  0.1393271   1.200  0.22999
## jobtechnician -0.1480734  0.1053715  -1.405  0.15995
## maritaldivorced -0.0652516  0.0654482  -0.997  0.31877
## maritalmarried -0.2709712  0.0448853  -6.037 1.57e-09 ***
## educationprimary -0.0911460  0.1041466  -0.875  0.38148
## educationsecondary 0.0308885  0.0925202   0.334  0.73849
## educationtertiary 0.2241288  0.0970079   2.310  0.02087 *
## in_defaultTRUE  0.0812191  0.1568900   0.518  0.60468
```

```
## housing_loanTRUE      -0.6613355  0.0384660 -17.193 < 2e-16 ***
## personal_loanTRUE     -0.3790967  0.0577740  -6.562 5.32e-11 ***
## campaign              -0.0882262  0.0091542  -9.638 < 2e-16 ***
## pdays                  0.0002633  0.0003030   0.869 0.38482
## previous               0.0089215  0.0061522   1.450 0.14703
## poutcomefailure       -1.4753805  1.0134828  -1.456 0.14546
## poutcomeother         -1.1920185  1.0149333  -1.174 0.24020
## poutcomesuccess        0.6866201  1.0145049   0.677 0.49853
## age_cathigh            0.8301434  0.1503614   5.521 3.37e-08 ***
## age_catmid            -0.6345220  0.0933299  -6.799 1.06e-11 ***
## balance_above_250TRUE  0.3514268  0.0392778   8.947 < 2e-16 ***
## was_contactedTRUE      1.7301365  1.0155648   1.704 0.08845 .
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## (Dispersion parameter for binomial family taken to be 1)
```

```
##      Null deviance: 25938  on 35938  degrees of freedom
```

```
## Residual deviance: 22439  on 35909  degrees of freedom
```

```
## AIC: 22499
```

```
##
```

```
## Number of Fisher Scoring iterations: 5
```

```
cat("\nModel 0 AUC:", round(results_1$auc, 4), "\n")
```

```
##
```

```
## Model 0 AUC: 0.7501
```

```
results_2 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
cat("Model 0 (Full) Summary:\n")
```

```
## Model 0 (Full) Summary:
```

```
print(summary(results_2$model))
```

```
##
```

```
## Call:
```

```
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -6.406e-01  1.707e-01  -3.752 0.000175 ***
## age          -6.836e-03  2.362e-03  -2.894 0.003802 **
## jobadmin      -3.016e-02  1.086e-01  -0.278 0.781126
## jobblue-collar -2.746e-01  1.071e-01  -2.565 0.010307 *
## jobentrepreneur -2.882e-01  1.458e-01  -1.977 0.048056 *
## jobhousemaid   -4.246e-01  1.535e-01  -2.766 0.005668 **
## jobmanagement  -8.174e-02  1.060e-01  -0.771 0.440509
## jobretired     -4.233e-02  1.317e-01  -0.321 0.747967
## jobself-employed -1.478e-01  1.342e-01  -1.101 0.270690
## jobservices    -2.264e-01  1.158e-01  -1.955 0.050581 .
## jobstudent     1.561e-01  1.389e-01   1.124 0.261219
```

```
## jobtechnician      -1.635e-01  1.051e-01  -1.556  0.119632
## maritaldivorced    -8.698e-02  6.532e-02  -1.332  0.182974
## maritalmarried     -2.758e-01  4.481e-02  -6.155  7.51e-10 ***
## educationprimary   -1.077e-01  1.040e-01  -1.036  0.300189
## educationsecondary  2.008e-02  9.240e-02   0.217  0.827922
## educationtertiary   2.153e-01  9.695e-02   2.221  0.026350 *
## in_defaultTRUE     -7.046e-02  1.556e-01  -0.453  0.650601
## balance            1.950e-05  4.896e-06   3.983  6.80e-05 ***
## housing_loanTRUE   -6.563e-01  3.845e-02 -17.067  < 2e-16 ***
## personal_loanTRUE  -4.027e-01  5.768e-02  -6.981  2.92e-12 ***
## campaign           -8.880e-02  9.125e-03  -9.731  < 2e-16 ***
## pdays              2.337e-04  3.028e-04   0.772  0.440211
## previous           9.164e-03  6.126e-03   1.496  0.134686
## poutcomefailure    -1.410e+00  1.005e+00  -1.403  0.160691
## poutcomeother      -1.126e+00  1.007e+00  -1.119  0.263275
## pcomesuccess        7.653e-01  1.006e+00   0.761  0.446854
## age_cathigh         8.329e-01  1.500e-01   5.552  2.83e-08 ***
## age_catmid          -6.397e-01  9.295e-02  -6.883  5.87e-12 ***
## was_contactedTRUE   1.693e+00  1.007e+00   1.681  0.092857 .
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## (Dispersion parameter for binomial family taken to be 1)
```

```
## Null deviance: 25938 on 35938 degrees of freedom
```

```
## Residual deviance: 22507 on 35909 degrees of freedom
```

```
## AIC: 22567
```

```
##
```

```
## Number of Fisher Scoring iterations: 5
```

```
cat("\nModel 0 AUC:", round(results_2$auc, 4), "\n")
```

```
##
```

```
## Model 0 AUC: 0.7473
```

```
results_3 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
cat("Model 0 (Full) Summary:\n")
```

```
## Model 0 (Full) Summary:
```

```
print(summary(results_3$model))
```

```
##
```

```
## Call:
```

```
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.6134904  0.1709913  -3.588  0.000333 ***
## age          -0.0079179  0.0023672  -3.345  0.000823 ***
## jobadmin      -0.0131042  0.1086954  -0.121  0.904041
## jobblue-collar -0.2602005  0.1072019  -2.427  0.015216 *
```

```
## jobentrepreneur      -0.2665517  0.1460310  -1.825  0.067954 .
## jobhousemaid         -0.4089310  0.1534656  -2.665  0.007707 **
## jobmanagement        -0.0777383  0.1061193  -0.733  0.463829
## jobretired           -0.0320104  0.1319485  -0.243  0.808317
## jobself-employed     -0.1499337  0.1343592  -1.116  0.264458
## jobservices          -0.2029607  0.1159668  -1.750  0.080090 .
## jobstudent           0.1494166  0.1390946   1.074  0.282729
## jobtechnician        -0.1506595  0.1052363  -1.432  0.152250
## maritaldivorced      -0.0585125  0.0654958  -0.893  0.371655
## maritalmarried       -0.2717694  0.0448979  -6.053  1.42e-09 ***
## educationprimary     -0.0975009  0.1041361  -0.936  0.349127
## educationsecondary   0.0304093  0.0925363   0.329  0.742442
## educationtertiary    0.2119541  0.0970542   2.184  0.028972 *
## in_defaultTRUE       0.1352259  0.1576108   0.858  0.390907
## housing_loanTRUE     -0.6467658  0.0384853 -16.806 < 2e-16 ***
## personal_loanTRUE    -0.3589100  0.0579535  -6.193  5.90e-10 ***
## campaign             -0.0875891  0.0091377  -9.585 < 2e-16 ***
## pdays                0.0003122  0.0003027   1.031  0.302352
## previous             0.0089816  0.0060932   1.474  0.140475
## poutcomefailure      -1.4343963  1.0070691  -1.424  0.154352
## poutcomeother        -1.1450443  1.0085277  -1.135  0.256224
## pcomesuccess         0.7345520  1.0080906   0.729  0.466212
## age_cathigh          0.8091695  0.1502594   5.385  7.24e-08 ***
## age_catmid           -0.6442918  0.0931151  -6.919  4.54e-12 ***
## trans_balance        0.1811894  0.0188453   9.615 < 2e-16 ***
## was_contactedTRUE    1.6755311  1.0091858   1.660  0.096858 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 25938 on 35938 degrees of freedom
## Residual deviance: 22429 on 35909 degrees of freedom
## AIC: 22489
##
## Number of Fisher Scoring iterations: 5
cat("\nModel 0 AUC:", round(results_3$auc, 4), "\n")
##
## Model 0 AUC: 0.7527
```

As it turns out, the model with `trans_balance` provided the largest AUC value. Let's do the same for the age variable.

```
results_4 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con
## Setting levels: control = FALSE, case = TRUE
## Setting direction: controls < cases
cat("Model 0 (Full) Summary:\n")
## Model 0 (Full) Summary:
print(summary(results_4$model))
##
```

```

## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.4502174  0.1572449  -9.223 < 2e-16 ***
## age           -0.0003739  0.0021412  -0.175  0.86136
## jobadmin       0.0129193  0.1087320   0.119  0.90542
## jobblue-collar -0.2329692  0.1071803  -2.174  0.02973 *
## jobentrepreneur -0.2455178  0.1454799  -1.688  0.09148 .
## jobhousemaid   -0.3376306  0.1517558  -2.225  0.02609 *
## jobmanagement -0.0620257  0.1060659  -0.585  0.55869
## jobretired     0.5412546  0.1204113   4.495 6.96e-06 ***
## jobself-employed -0.1156630  0.1338724  -0.864  0.38760
## jobservices    -0.1707397  0.1159673  -1.472  0.14094
## jobstudent     0.4777284  0.1341782   3.560 0.00037 ***
## jobtechnician  -0.1366726  0.1052907  -1.298  0.19427
## maritaldivorced -0.1616900  0.0646245  -2.502  0.01235 *
## maritalmarried  -0.3482379  0.0444087  -7.842 4.45e-15 ***
## educationprimary -0.1111322  0.1024709  -1.085  0.27813
## educationsecondary 0.0103142  0.0911723   0.113  0.90993
## educationtertiary 0.1872918  0.0955655   1.960  0.05002 .
## in_defaultTRUE   0.1138231  0.1576894   0.722  0.47041
## housing_loanTRUE -0.6692550  0.0381131 -17.560 < 2e-16 ***
## personal_loanTRUE -0.4000360  0.0577399  -6.928 4.26e-12 ***
## campaign        -0.0927044  0.0091903 -10.087 < 2e-16 ***
## pdays          0.0001728  0.0003007   0.575  0.56552
## previous        0.0089131  0.0060803   1.466  0.14268
## poutcomefailure -1.6156826  1.0122552  -1.596  0.11046
## poutcomeother   -1.3079241  1.0136927  -1.290  0.19696
## pcomesuccess     0.5717598  1.0132230   0.564  0.57255
## trans_balance    0.1850167  0.0186595   9.915 < 2e-16 ***
## was_contactedTRUE 1.9080475  1.0143339   1.881  0.05996 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 25938  on 35938  degrees of freedom
## Residual deviance: 22673  on 35911  degrees of freedom
## AIC: 22729
##
## Number of Fisher Scoring iterations: 5
cat("\nModel 0 AUC:", round(results_4$auc, 4), "\n")

##
## Model 0 AUC: 0.7469
results_5 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con

## Setting levels: control = FALSE, case = TRUE
## Setting direction: controls < cases

```

```

cat("Model 0 (Full) Summary:\n")

## Model 0 (Full) Summary:
print(summary(results_5$model))

##
## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.8374123   0.1574672  -5.318 1.05e-07 ***
## jobadmin       -0.0075029   0.1086820  -0.069  0.94496
## jobblue-collar -0.2495943   0.1071474  -2.329  0.01984 *
## jobentrepreneur -0.2781073   0.1460377  -1.904  0.05686 .
## jobhousemaid   -0.4308633   0.1533645  -2.809  0.00496 **
## jobmanagement -0.0839894   0.1060805  -0.792  0.42851
## jobretired     -0.1132237   0.1293647  -0.875  0.38145
## jobself-employed -0.1469341   0.1342798  -1.094  0.27385
## jobservices    -0.1921075   0.1159075  -1.657  0.09743 .
## jobstudent      0.1908937   0.1386650   1.377  0.16862
## jobtechnician  -0.1467205   0.1052314  -1.394  0.16324
## maritaldivorced -0.1330814   0.0616205  -2.160  0.03080 *
## maritalmarried  -0.3281792   0.0416790  -7.874 3.44e-15 ***
## educationprimary -0.0968258   0.1040360  -0.931  0.35201
## educationsecondary 0.0525773   0.0922174   0.570  0.56858
## educationtertiary 0.2467997   0.0964129   2.560  0.01047 *
## in_defaultTRUE   0.1363193   0.1575837   0.865  0.38701
## housing_loanTRUE -0.6322321   0.0382188 -16.542 < 2e-16 ***
## personal_loanTRUE -0.3589751   0.0579549  -6.194 5.86e-10 ***
## campaign        -0.0884619   0.0091502  -9.668 < 2e-16 ***
## pdays           0.0003234   0.0003026   1.069  0.28511
## previous        0.0088964   0.0060810   1.463  0.14347
## poutcomefailure -1.4868932   1.0064765  -1.477  0.13959
## poutcomeother   -1.1922040   1.0079584  -1.183  0.23689
## pcomesuccess     0.6811735   1.0074839   0.676  0.49897
## age_catighigh    0.5895623   0.1350706   4.365 1.27e-05 ***
## age_catemid     -0.7267423   0.0900203  -8.073 6.85e-16 ***
## trans_balance    0.1762214   0.0187545   9.396 < 2e-16 ***
## was_contactedTRUE 1.7255784   1.0086025   1.711  0.08711 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 25938  on 35938  degrees of freedom
## Residual deviance: 22440  on 35910  degrees of freedom
## AIC: 22498
##
## Number of Fisher Scoring iterations: 5
cat("\nModel 0 AUC:", round(results_5$auc, 4), "\n")

##

```

```
## Model 0 AUC: 0.7527
```

By dropping the continuous age variable, the AUC value increased slightly. We will keep `age_categ` and `trans_balance` in our model.

```
results_6 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
cat("Model 0 (Full) Summary:\n")
```

```
## Model 0 (Full) Summary:
```

```
print(summary(results_6$model))
```

```
##
```

```
## Call:
```

```
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.8376127  0.1574633  -5.319 1.04e-07 ***
## jobadmin      -0.0066671  0.1086750  -0.061  0.95108
## jobblue-collar -0.2496748  0.1071464  -2.330  0.01979 *
## jobentrepreneur -0.2782280  0.1460376  -1.905  0.05676 .
## jobhousemaid   -0.4307318  0.1533627  -2.809  0.00498 **
## jobmanagement -0.0832705  0.1060798  -0.785  0.43247
## jobretired     -0.1123465  0.1293453  -0.869  0.38508
## jobself-employed -0.1469464  0.1342770  -1.094  0.27380
## jobservices    -0.1922737  0.1159065  -1.659  0.09714 .
## jobstudent      0.1912763  0.1386593   1.379  0.16775
## jobtechnician  -0.1468056  0.1052301  -1.395  0.16299
## maritaldivorced -0.1336568  0.0616192  -2.169  0.03008 *
## maritalmarried -0.3279476  0.0416748  -7.869 3.57e-15 ***
## educationprimary -0.0970837  0.1040373  -0.933  0.35074
## educationsecondary 0.0530627  0.0922168   0.575  0.56501
## educationtertiary  0.2473966  0.0964127   2.566  0.01029 *
## in_defaultTRUE    0.1356904  0.1575822   0.861  0.38920
## housing_loanTRUE  -0.6327825  0.0382202 -16.556 < 2e-16 ***
## personal_loanTRUE -0.3581976  0.0579417  -6.182 6.33e-10 ***
## campaign         -0.0884888  0.0091495  -9.671 < 2e-16 ***
## pdays           0.0003542  0.0003022   1.172  0.24108
## previous         0.0090885  0.0061065   1.488  0.13667
## poutcomefailure  0.2303562  0.0902870   2.551  0.01073 *
## poutcomeother    0.5253152  0.1036375   5.069 4.00e-07 ***
## pcomesuccess     2.4003142  0.0857819  27.982 < 2e-16 ***
## age_categhigh    0.5914810  0.1350493   4.380 1.19e-05 ***
## age_categmid     -0.7265676  0.0900140  -8.072 6.93e-16 ***
## trans_balance    0.1762001  0.0187523   9.396 < 2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
```

```
##      Null deviance: 25938  on 35938  degrees of freedom
```

```

## Residual deviance: 22443 on 35911 degrees of freedom
## AIC: 22499
##
## Number of Fisher Scoring iterations: 5
cat("\nModel 0 AUC:", round(results_6$auc, 4), "\n")

##
## Model 0 AUC: 0.7528
results_7 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con

## Setting levels: control = FALSE, case = TRUE
## Setting direction: controls < cases
cat("Model 0 (Full) Summary:\n")

## Model 0 (Full) Summary:
print(summary(results_7$model))

##
## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.8408944  0.1574679  -5.340 9.29e-08 ***
## jobadmin      -0.0068580  0.1086844  -0.063  0.94969
## jobblue-collar -0.2499472  0.1071513  -2.333  0.01967 *
## jobentrepreneur -0.2776410  0.1460391  -1.901  0.05728 .
## jobhousemaid  -0.4308301  0.1533684  -2.809  0.00497 **
## jobmanagement -0.0827633  0.1060806  -0.780  0.43528
## jobretired    -0.1136540  0.1293716  -0.879  0.37967
## jobself-employed -0.1463413  0.1342856  -1.090  0.27581
## jobservices   -0.1924512  0.1159149  -1.660  0.09686 .
## jobstudent     0.1921216  0.1386622   1.386  0.16589
## jobtechnician -0.1463433  0.1052371  -1.391  0.16435
## maritaldivorced -0.1336270  0.0616238  -2.168  0.03013 *
## maritalmarried -0.3279805  0.0416769  -7.870 3.56e-15 ***
## educationprimary -0.0953683  0.1040357  -0.917  0.35931
## educationsecondary 0.0531706  0.0922253   0.577  0.56426
## educationtertiary 0.2475252  0.0964173   2.567  0.01025 *
## in_defaultTRUE   0.1364858  0.1575885   0.866  0.38644
## housing_loanTRUE -0.6313824  0.0382115 -16.523 < 2e-16 ***
## personal_loanTRUE -0.3584897  0.0579558  -6.186 6.19e-10 ***
## campaign        -0.0880587  0.0091365  -9.638 < 2e-16 ***
## pdays           0.0003151  0.0003028   1.041  0.29804
## poutcomefailure -1.4940229  1.0076110  -1.483  0.13814
## poutcomeother   -1.1885833  1.0090949  -1.178  0.23885
## poutcomesuccess  0.6755744  1.0086206   0.670  0.50299
## age_categhigh    0.5912829  0.1350724   4.378 1.20e-05 ***
## age_categmid     -0.7257887  0.0900274  -8.062 7.51e-16 ***
## trans_balance    0.1763103  0.0187532   9.402 < 2e-16 ***
## was_contactedTRUE 1.7605128  1.0094675   1.744  0.08116 .
## ---

```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 25938  on 35938  degrees of freedom
## Residual deviance: 22442  on 35911  degrees of freedom
## AIC: 22498
##
## Number of Fisher Scoring iterations: 5
```

```
cat("\nModel 0 AUC:", round(results_7$auc, 4), "\n")
```

```
##
## Model 0 AUC: 0.7527
```

Neither of the previous variable variations are statistically significant. We will drop them.

```
results_7 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
cat("Model 0 (Full) Summary:\n")
```

```
## Model 0 (Full) Summary:
```

```
print(summary(results_7$model))
```

```
##
## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.8411788   0.1574642  -5.342 9.19e-08 ***
## jobadmin       -0.0059948   0.1086773  -0.055  0.95601
## jobblue-collar -0.2500357   0.1071503  -2.334  0.01962 *
## jobentrepreneur -0.2777613   0.1460390  -1.902  0.05718 .
## jobhousemaid   -0.4306978   0.1533667  -2.808  0.00498 **
## jobmanagement  -0.0820266   0.1060801  -0.773  0.43937
## jobretired     -0.1127691   0.1293519  -0.872  0.38332
## jobself-employed -0.1463498   0.1342828  -1.090  0.27577
## jobservices    -0.1926289   0.1159139  -1.662  0.09655 .
## jobstudent      0.1925315   0.1386562   1.389  0.16497
## jobtechnician  -0.1464246   0.1052359  -1.391  0.16411
## maritaldivorced -0.1342247   0.0616226  -2.178  0.02939 *
## maritalmarried  -0.3277454   0.0416726  -7.865 3.70e-15 ***
## educationprimary -0.0956007   0.1040370  -0.919  0.35814
## educationsecondary 0.0536720   0.0922249   0.582  0.56059
## educationtertiary 0.2481570   0.0964172   2.574  0.01006 *
## in_defaultTRUE   0.1358531   0.1575871   0.862  0.38864
## housing_loanTRUE -0.6319105   0.0382129 -16.537 < 2e-16 ***
## personal_loanTRUE -0.3576760   0.0579420  -6.173 6.70e-10 ***
## campaign        -0.0880767   0.0091355  -9.641 < 2e-16 ***
## pdays           0.0003460   0.0003024   1.144  0.25258
## poutcomefailure  0.2586944   0.0882979   2.930  0.00339 **
## poutcomeother    0.5646544   0.1001056   5.641 1.69e-08 ***
```

```
## poutcomesuccess      2.4302278  0.0834615  29.118 < 2e-16 ***
## age_categhigh        0.5932798  0.1350505   4.393 1.12e-05 ***
## age_categmid        -0.7255888  0.0900212  -8.060 7.62e-16 ***
## trans_balance        0.1762924  0.0187509   9.402 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 25938  on 35938  degrees of freedom
## Residual deviance: 22445  on 35912  degrees of freedom
## AIC: 22499
##
## Number of Fisher Scoring iterations: 5
```

```
cat("\nModel 0 AUC:", round(results_7$auc, 4), "\n")
```

```
##
## Model 0 AUC: 0.7528
```

We can also drop the pdays and in_default variables as it is not statistically significant.

```
results_8 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
cat("Model 0 (Full) Summary:\n")
```

```
## Model 0 (Full) Summary:
```

```
print(summary(results_8$model))
```

```
##
## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.842487   0.157425  -5.352 8.72e-08 ***
## jobadmin      -0.007279   0.108661  -0.067  0.94659
## jobblue-collar -0.248700   0.107133  -2.321  0.02026 *
## jobentrepreneur -0.276988   0.146010  -1.897  0.05782 .
## jobhousemaid   -0.432732   0.153388  -2.821  0.00478 **
## jobmanagement  -0.082353   0.106072  -0.776  0.43752
## jobretired     -0.113783   0.129362  -0.880  0.37909
## jobself-employed -0.147437   0.134293  -1.098  0.27226
## jobservices    -0.192592   0.115893  -1.662  0.09655 .
## jobstudent      0.189709   0.138648   1.368  0.17123
## jobtechnician  -0.147927   0.105221  -1.406  0.15976
## maritaldivorced -0.133751   0.061617  -2.171  0.02995 *
## maritalmarried  -0.329202   0.041662  -7.902 2.75e-15 ***
## educationprimary -0.094823   0.104043  -0.911  0.36210
## educationsecondary 0.054332   0.092232   0.589  0.55581
## educationtertiary 0.246603   0.096418   2.558  0.01054 *
## housing_loanTRUE -0.627170   0.037916 -16.541 < 2e-16 ***
## personal_loanTRUE -0.355290   0.057840  -6.143 8.12e-10 ***
```

```
## campaign          -0.087894    0.009132   -9.625 < 2e-16 ***
## poutcomefailure    0.337355    0.054197    6.225 4.83e-10 ***
## poutcomeother      0.637612    0.076174    8.371 < 2e-16 ***
## poutcomesuccess    2.487067    0.066429   37.440 < 2e-16 ***
## age_categhigh      0.592646    0.135073    4.388 1.15e-05 ***
## age_categmid       -0.722988    0.090027   -8.031 9.69e-16 ***
## trans_balance      0.173149    0.018527    9.346 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 25938  on 35938  degrees of freedom
## Residual deviance: 22447  on 35914  degrees of freedom
## AIC: 22497
##
## Number of Fisher Scoring iterations: 5
```

```
cat("\nModel 0 AUC:", round(results_8$auc, 4), "\n")
```

```
##
## Model 0 AUC: 0.7527
```

What we are left with is a model that mostly has statistically significant variables. Yet, we still the `education` and `'job'` variable that have categories that are not significant. We should try grouping into larger groups.

```
bank_data <- bank_data %>%
  mutate(has_higher_education = ifelse(education == "tertiary", TRUE, FALSE))
```

```
results_9 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
cat("Model 0 (Full) Summary:\n")
```

```
## Model 0 (Full) Summary:
```

```
print(summary(results_9$model))
```

```
##
## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.815303   0.130826  -6.232 4.61e-10 ***
## jobadmin        0.015484   0.108201   0.143  0.88621
## jobblue-collar  -0.270612   0.106690  -2.536  0.01120 *
## jobentrepreneur -0.274277   0.145829  -1.881  0.06000 .
## jobhousemaid    -0.471405   0.152409  -3.093  0.00198 **
## jobmanagement  -0.075305   0.105812  -0.712  0.47666
## jobretired      -0.132253   0.129092  -1.024  0.30560
## jobself-employed -0.139755   0.134140  -1.042  0.29748
## jobservices     -0.173448   0.115566  -1.501  0.13339
## jobstudent       0.195395   0.137863   1.417  0.15639
## jobtechnician   -0.126411   0.104781  -1.206  0.22765
```

```
## maritaldivorced      -0.139423    0.061581   -2.264  0.02357 *
## maritalmarried      -0.335485    0.041601   -8.064  7.37e-16 ***
## housing_loanTRUE    -0.625183    0.037885  -16.502 < 2e-16 ***
## personal_loanTRUE   -0.351673    0.057784   -6.086  1.16e-09 ***
## campaign            -0.088076    0.009133   -9.644 < 2e-16 ***
## poutcomefailure      0.339913    0.054180    6.274  3.52e-10 ***
## poutcomeother       0.638222    0.076204    8.375 < 2e-16 ***
## pcomesuccess        2.492507    0.066400   37.538 < 2e-16 ***
## age_cathigh         0.576945    0.134896    4.277  1.89e-05 ***
## age_catmid          -0.727628    0.090085   -8.077  6.63e-16 ***
## trans_balance       0.172873    0.018526    9.331 < 2e-16 ***
## has_higher_educationTRUE 0.218623    0.049117    4.451  8.54e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 25938  on 35938  degrees of freedom
## Residual deviance: 22453  on 35916  degrees of freedom
## AIC: 22499
##
## Number of Fisher Scoring iterations: 5
cat("\nModel 0 AUC:", round(results_9$auc, 4), "\n")
```

```
##
## Model 0 AUC: 0.7516
```

We can try grouping the jobs by their likelihood of subscription.

```
bank_data <- bank_data %>%
  mutate(job_type = case_when(
    job %in% c("retired", "student") ~ "high_prob",
    job %in% c("management", "admin", "unemployed", "self-employed") ~ "mid_prob",
    job %in% c("blue-collar", "entrepreneur", "housemaid", "services", "technician") ~ "low_prob"
  ),
  job_type = factor(job_type, levels = c("low_prob", "mid_prob", "high_prob")),
  job_type = relevel(job_type, ref = "low_prob"))
```

```
results_10 <- train_evaluate_model(bank_data[, !names(bank_data) %in% c("day", "month", "duration", "con")])
```

```
## Setting levels: control = FALSE, case = TRUE
## Setting direction: controls < cases
```

```
cat("Model 0 (Full) Summary:\n")
```

```
## Model 0 (Full) Summary:
```

```
print(summary(results_10$model))
```

```
##
## Call:
## glm(formula = subscribed ~ ., family = binomial, data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.946985   0.088502 -10.700 < 2e-16 ***
```

```
## maritaldivorced      -0.166810    0.060980   -2.735  0.00623 **
## maritalmarried      -0.369372    0.040651   -9.086  < 2e-16 ***
## housing_loanTRUE    -0.626500    0.037438  -16.734  < 2e-16 ***
## personal_loanTRUE   -0.354787    0.057600   -6.160  7.30e-10 ***
## campaign            -0.088022    0.009129   -9.642  < 2e-16 ***
## poutcomefailure     0.346634    0.054121    6.405  1.51e-10 ***
## poutcomeother       0.649465    0.076074    8.537  < 2e-16 ***
## pcomesuccess        2.507839    0.066306   37.822  < 2e-16 ***
## age_cathigh         0.405157    0.115271    3.515  0.00044 ***
## age_catmid          -0.786470    0.085368   -9.213  < 2e-16 ***
## trans_balance       0.171898    0.018501    9.291  < 2e-16 ***
## has_higher_educationTRUE 0.199869    0.041578    4.807  1.53e-06 ***
## job_typemid_prob    0.170359    0.041718    4.084  4.43e-05 ***
## job_typehigh_prob   0.224367    0.069874    3.211  0.00132 **
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
```

```
## Null deviance: 25938 on 35938 degrees of freedom
```

```
## Residual deviance: 22472 on 35924 degrees of freedom
```

```
## AIC: 22502
```

```
##
```

```
## Number of Fisher Scoring iterations: 5
```

```
cat("\nModel 0 AUC:", round(results_10$auc, 4), "\n")
```

```
##
```

```
## Model 0 AUC: 0.749
```

Now we have a model where all variables are statistically significant. Let's train our model with k-fold cross-validation with k=10 to confirm the results. We will use upsampling of subscribers due to the imbalanced data set.

```
set.seed(1)
```

```
train_control <- trainControl(
  method = "cv",
  number = 10,
  savePredictions = "final",
  classProbs = TRUE,
  summaryFunction = multiClassSummary,
  sampling = "up"
)
```

```
bank_data$subscribed_f <- factor(bank_data$subscribed, levels = c(FALSE, TRUE), labels = c("No", "Yes"))
```

```
results_11 <- train(
  subscribed_f ~ .,
  data = bank_data[, !names(bank_data) %in% c("day", "month", "duration", "contact_type", "balance_above")]
  method = "glm",
  family = binomial,
  trControl = train_control,
  metric = "AUC"
)
```

```
print(results_11)
```

```
## Generalized Linear Model
##
## 44923 samples
##    9 predictor
##    2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 40432, 40431, 40431, 40430, 40430, 40430, ...
## Additional sampling using up-sampling
##
## Resampling results:
##
##   logLoss      AUC      prAUC      Accuracy      Kappa      F1      Sensitivity
##   0.5842027  0.7342799  0.6550979  0.7387085  0.2174986  0.8366532  0.7579405
##   Specificity Pos_Pred_Value Neg_Pred_Value Precision Recall
##   0.5935166   0.9336799      0.2452619      0.9336799  0.7579405
##   Detection_Rate Balanced_Accuracy
##   0.6692788      0.6757285
```

The cross-validated AUC value is 0,734 which is lower than 0,749 that we got when training and testing took place on a fixed data set split.

```
confusionMatrix(results_11$pred$pred, results_11$pred$obs)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##           No 30066 2136
##           Yes 9602 3119
##
##           Accuracy : 0.7387
##           95% CI : (0.7346, 0.7428)
##           No Information Rate : 0.883
##           P-Value [Acc > NIR] : 1
##
##           Kappa : 0.2175
##
## Mcnemar's Test P-Value : <2e-16
##
##           Sensitivity : 0.7579
##           Specificity : 0.5935
##           Pos Pred Value : 0.9337
##           Neg Pred Value : 0.2452
##           Prevalence : 0.8830
##           Detection Rate : 0.6693
##           Detection Prevalence : 0.7168
##           Balanced Accuracy : 0.6757
##
##           'Positive' Class : No
##
```

The sensitivity (the true positive rate) of the model is quite high. 75,73% clients who would subscribe to a deposit are being recognized as “subscribers”. Specificity (the true negative rate) is moderate (59,31%), meaning that the majority of clients who have subscribed are being classified as “subscribers”.

```
summary(results_11)
```

```
##
## Call:
## NULL
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.003899   0.043796  22.922 < 2e-16 ***
## maritaldivorced  -0.156189   0.027020  -5.781 7.45e-09 ***
## maritalmarried   -0.407518   0.017993 -22.649 < 2e-16 ***
## housing_loanTRUE -0.607570   0.016240 -37.412 < 2e-16 ***
## personal_loanTRUE -0.328000   0.023551 -13.927 < 2e-16 ***
## campaign         -0.083350   0.003523 -23.661 < 2e-16 ***
## poutcomefailure   0.360618   0.024189  14.908 < 2e-16 ***
## poutcomeother     0.583249   0.035958  16.220 < 2e-16 ***
## poutcomesuccess   2.577340   0.046503  55.424 < 2e-16 ***
## age_catelhigh     0.596389   0.062568   9.532 < 2e-16 ***
## age_catelmid      -0.720879   0.042856 -16.821 < 2e-16 ***
## trans_balance     0.204617   0.008224  24.880 < 2e-16 ***
## has_higher_educationTRUE 0.232914   0.018601  12.522 < 2e-16 ***
## job_typeelmid_prob 0.132183   0.018158   7.280 3.35e-13 ***
## job_typeelhigh_prob 0.171411   0.033368   5.137 2.79e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 109983  on 79335  degrees of freedom
## Residual deviance:  94442  on 79321  degrees of freedom
## AIC: 94472
##
## Number of Fisher Scoring iterations: 5
```

Conclusion

1. We have produced a logistic regression model with an accuracy score of 0,7381. The sensitivity is 0,7573 and the specificity is 0,5931.
2. Most important parameters for choosing a potential bank deposit subscriber is the outcome of a previous campaign and whether the client has borrowed a loan.
3. The call duration, day, month and contact type variables cannot be used in the model due to in not being available before calling the customer even though they might be significant determining features.
4. The researchers should look into gathering additional data during aforementioned days and months.
5. Additional descriptive client variables, such as previously used bank products and household income, would help increase model performance.